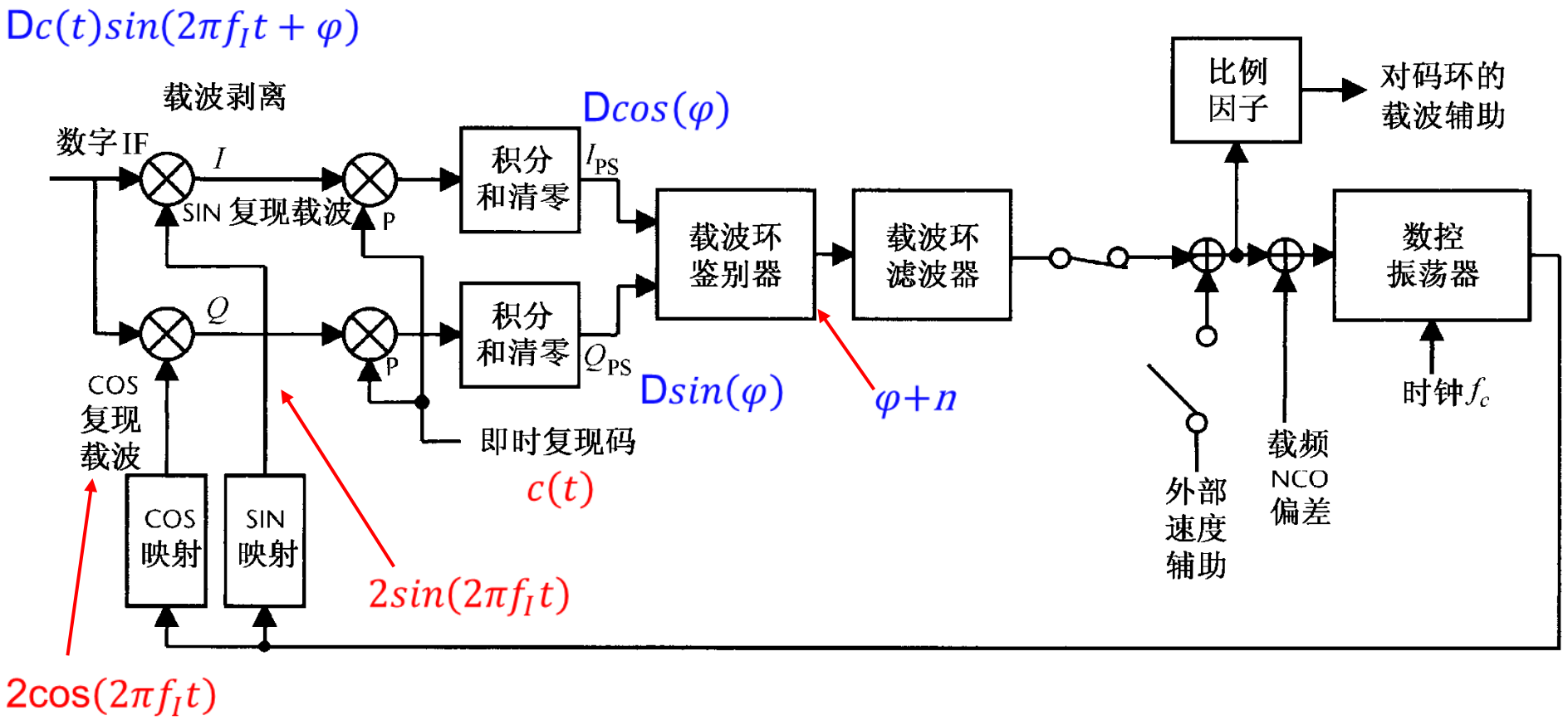
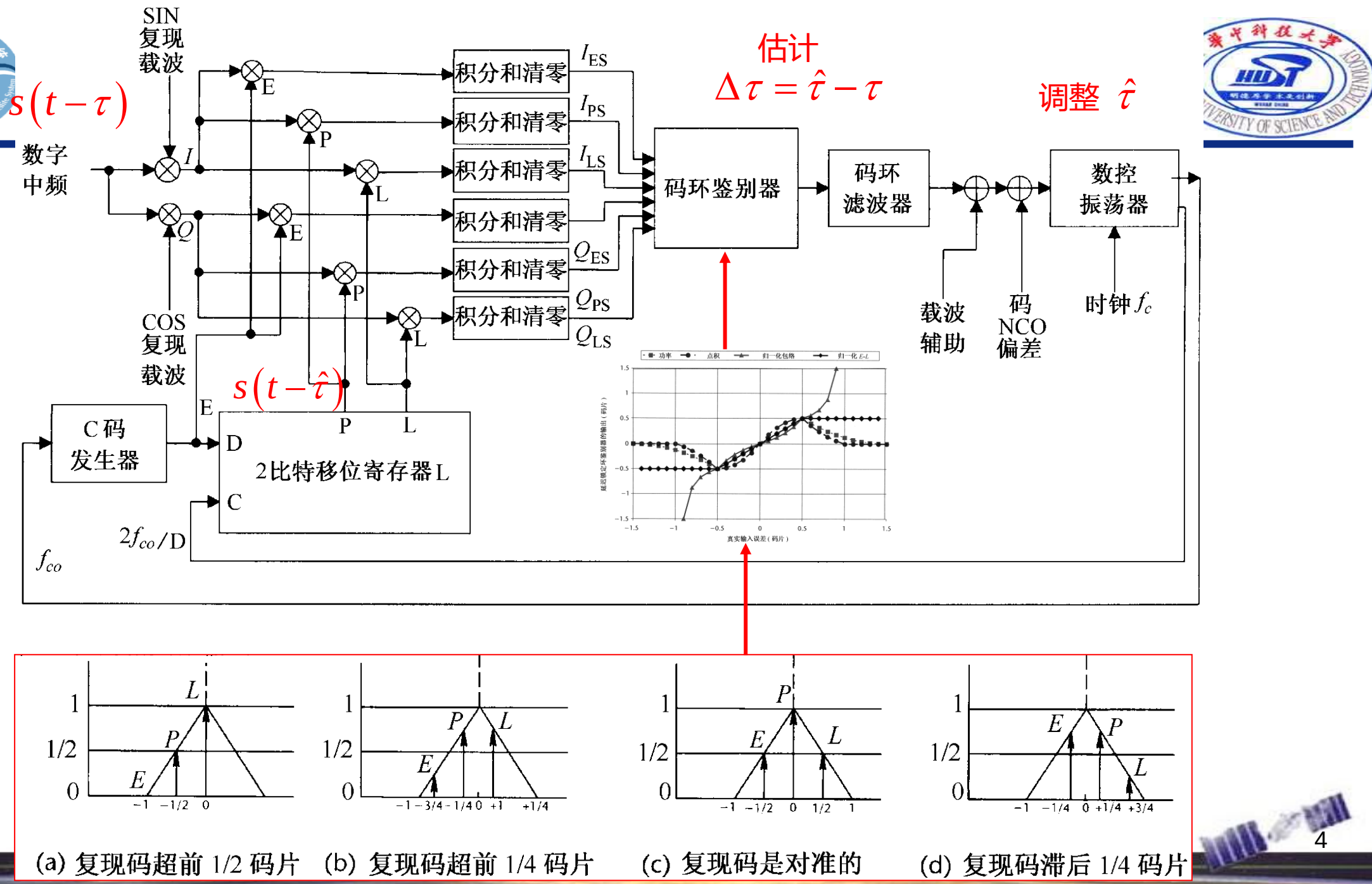
环路滤波器
——原理及仿真

主要内容

- 背景
- 工作原理
- 基于simulink的环路仿真
- 实验内容

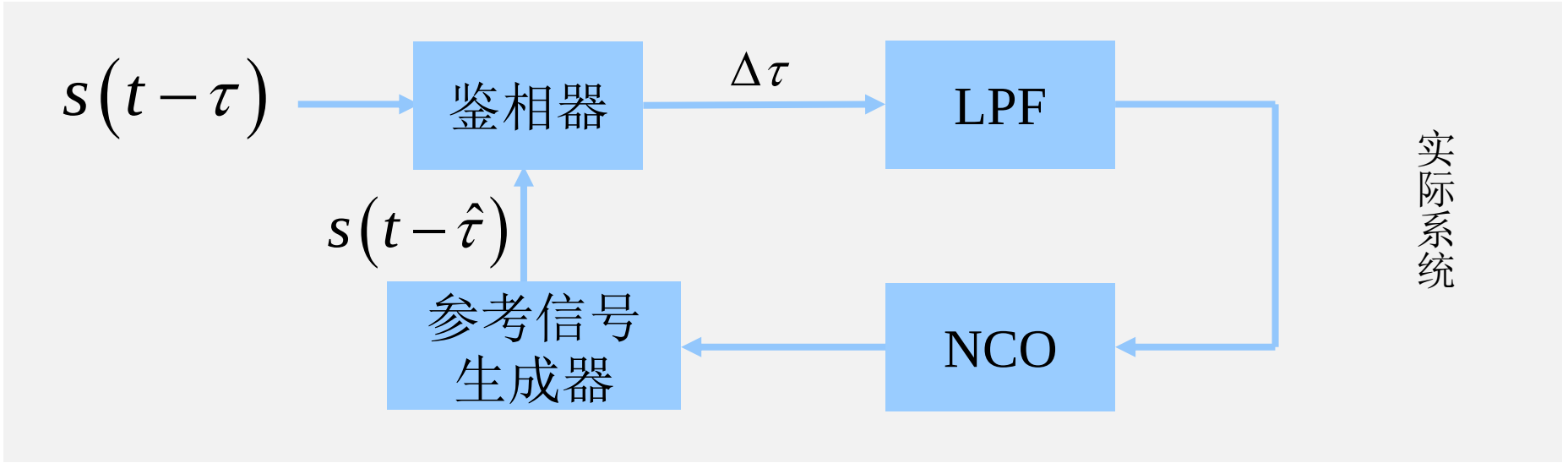
背景

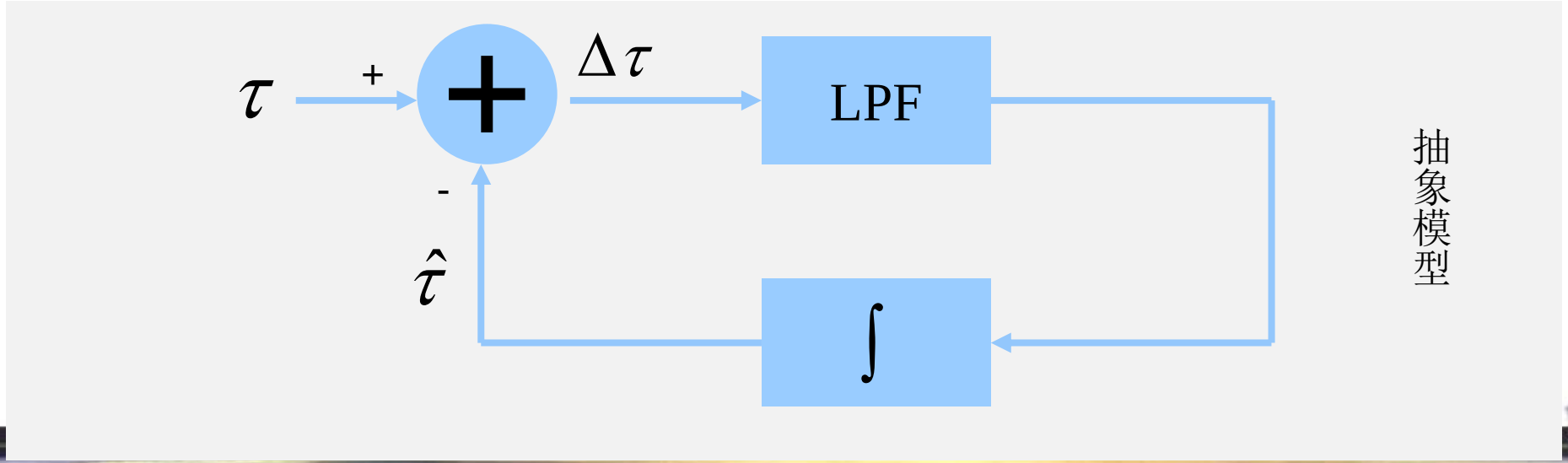

$Dc(t)sin(2\pi f_I t+\varphi)$
载波剥离
数字IF
I
SIN 复现载波
P
$Dcos(\varphi)$
积分和清零
I_{PS}
Q
P
积分和清零
Q_{PS}
$Dsin(\varphi)$
载波环鉴别器
载波环滤波器
$\varphi+n$
比例因子
对码环的载波辅助
数控振荡器
时钟f_c
COS复现载波
即时复现码
$c(t)$
COS映射
SIN映射
$2sin(2\pi f_I t)$
外部速度辅助
载频NCO偏差
$2cos(2\pi f_I t)$



(a) 复现码超前 1/2 码片　(b) 复现码超前 1/4 码片　(c) 复现码是对准的　(d) 复现码滞后 1/4 码片

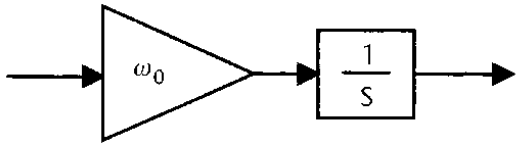
工作原理


$s(t-\tau)$
鉴相器
$\Delta\tau$
LPF
$s(t-\hat{\tau})$
参考信号
生成器
NCO
实际系统

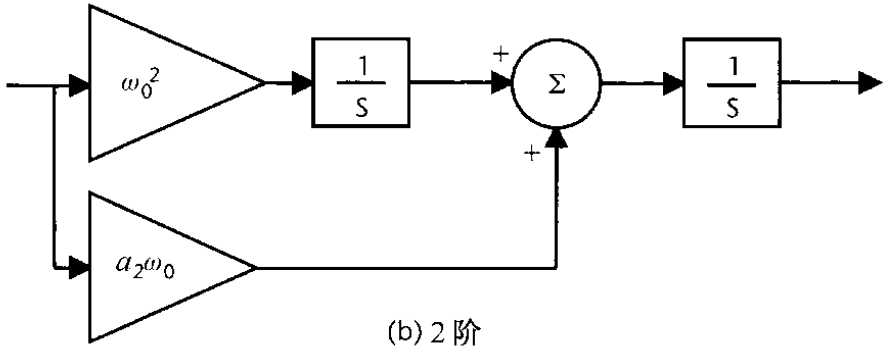

τ
+
$\Delta\tau$
LPF
-
$\hat{\tau}$
$\int$
抽象模型

工作原理

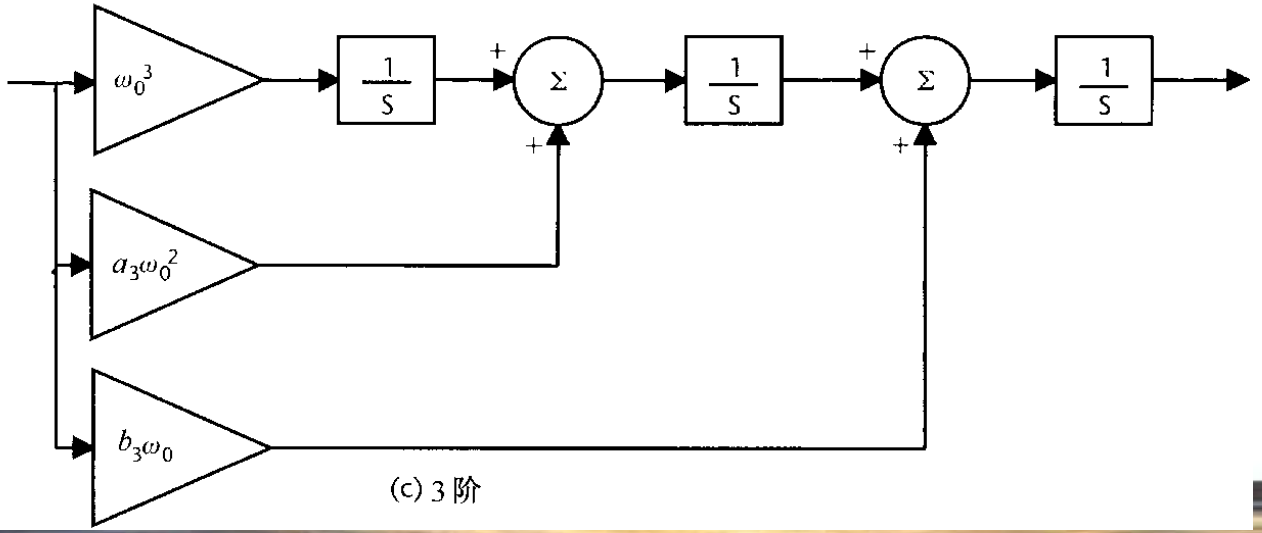
滤波器结构
（模拟）



(a) 1 阶



(b) 2 阶



(c) 3 阶

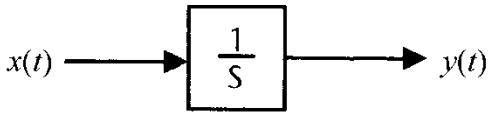
工作原理

环路特性

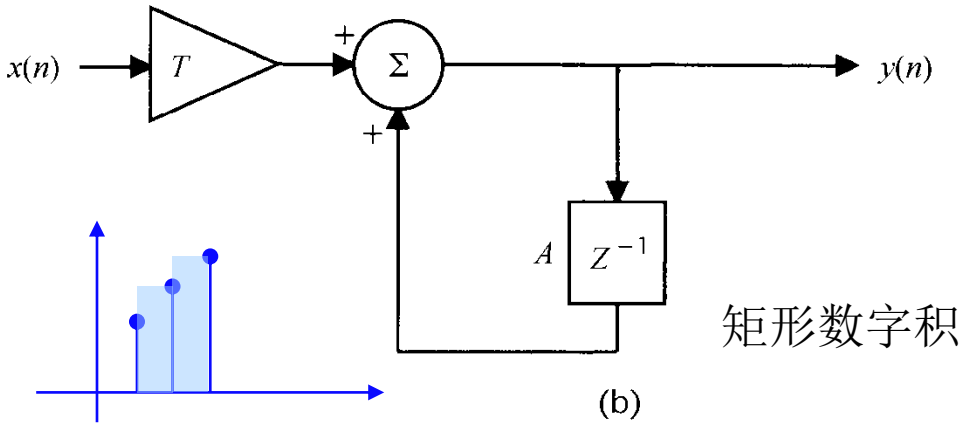
环路阶数	噪声带宽 Bn（Hz）	典型滤波器 参数	稳态误差
一阶	$\frac{\omega_0}{4}$	ω_0 $B_n = 0.25\,\omega_0$	$\frac{(\mathrm{d}R/\mathrm{d}t)}{\omega_0}$
二阶	$\frac{\omega_0(1+a_2^2)}{4a_2}$	ω_0^2 $a_2\omega_0 = 1.414\omega_0$ $B_n = 0.53\,\omega_0$	$\frac{(\mathrm{d}^2R/\mathrm{d}t^2)}{\omega_0^2}$
三阶	$\frac{\omega_0(a_3b_3^2+a_3^2-b_3)}{4(a_3b_3-1)}$	ω_0^3 $a_3\omega_0^2 = 1.1\omega_0^2$ $b_3\omega_0 = 2.4\omega_0$ $B_n = 0.7845\,\omega_0$	$\frac{(\mathrm{d}^3R/\mathrm{d}t^3)}{\omega_0^3}$

数字环路

模拟积分器



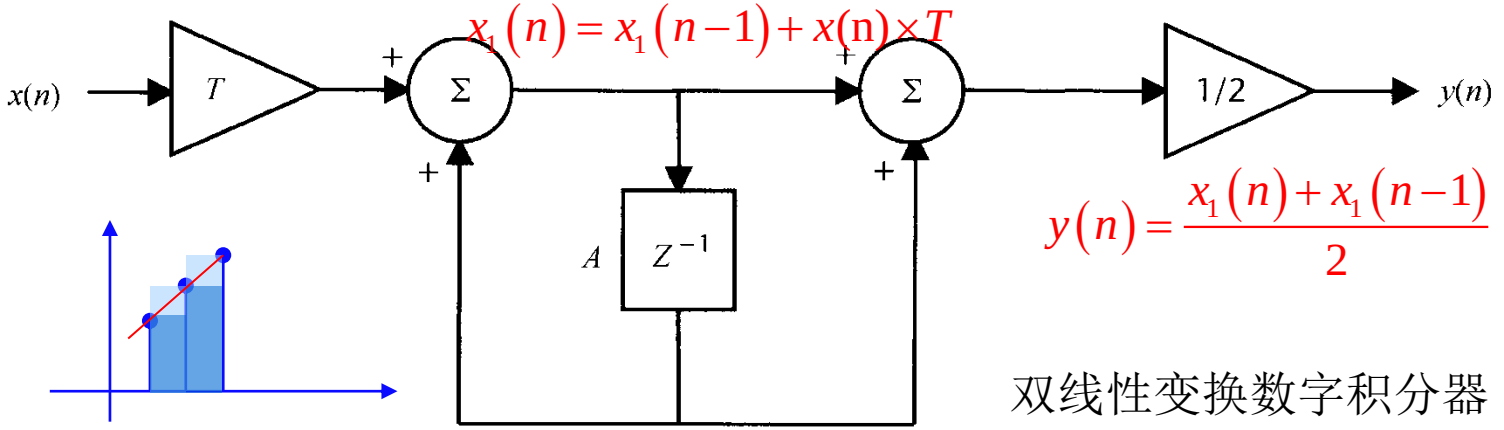
(a)



矩形数字积分器

(b)

$$x_1(n) = x_1(n-1) + x(n) \times T$$



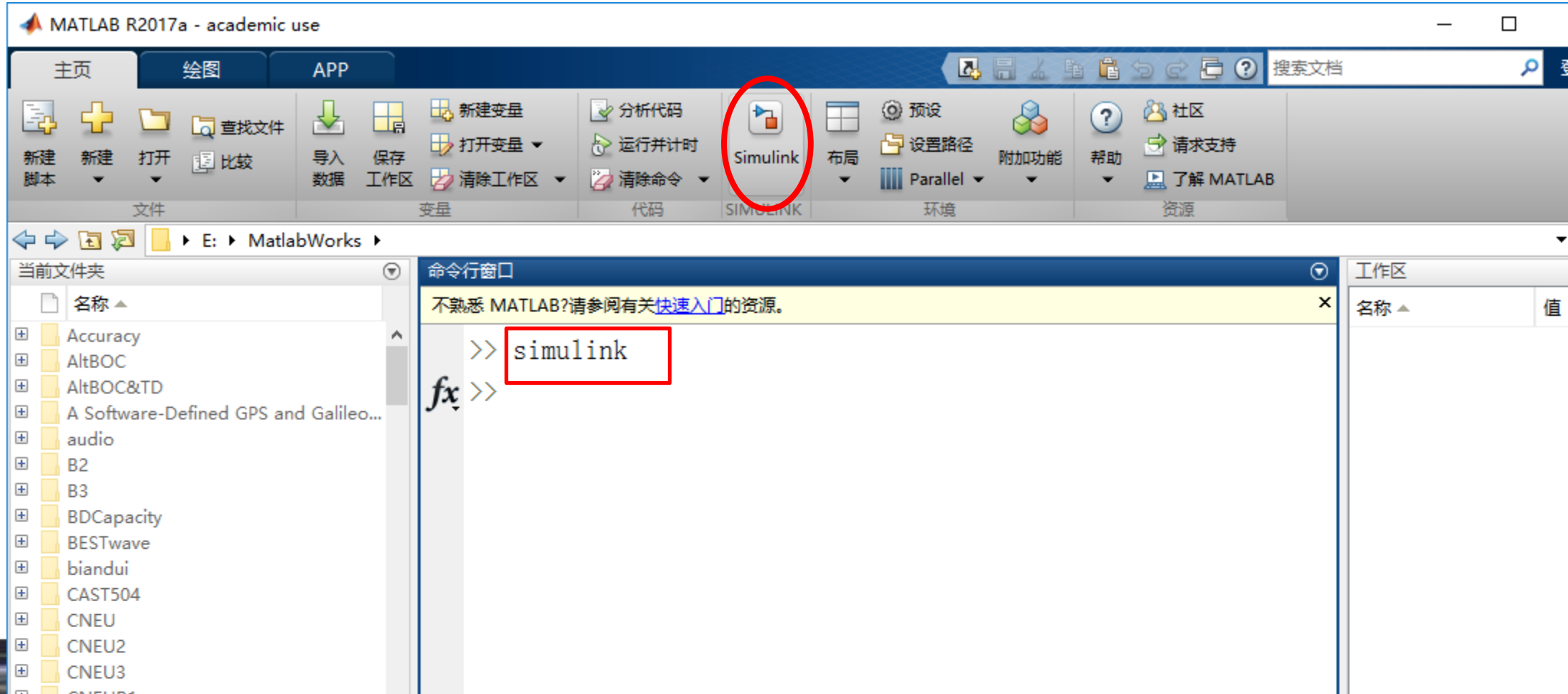
$$y(n) = \frac{x_1(n) + x_1(n-1)}{2}$$

双线性变换数字积分器

(c)

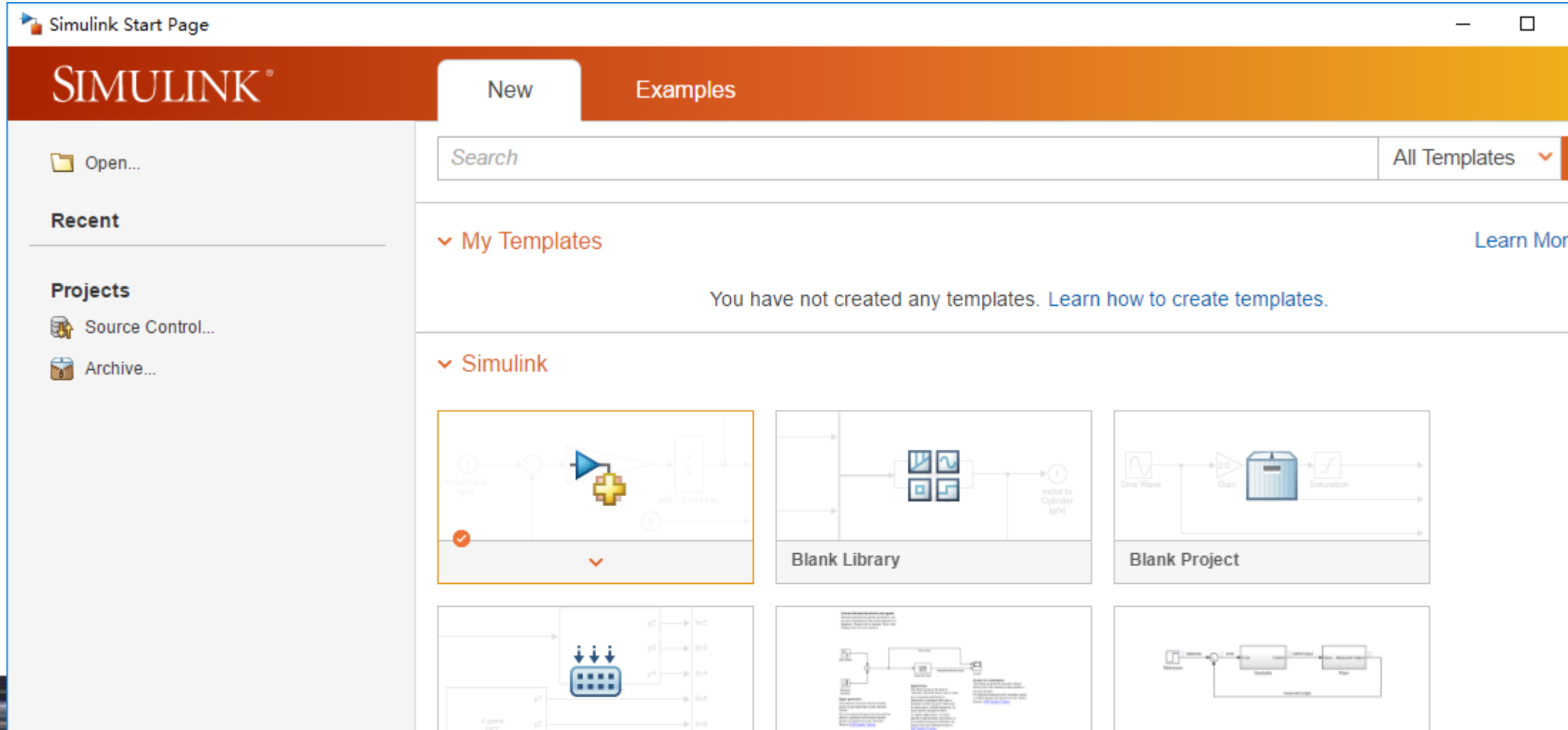
基于simulink的环路仿真

1. 在matlab界面中打开simulink



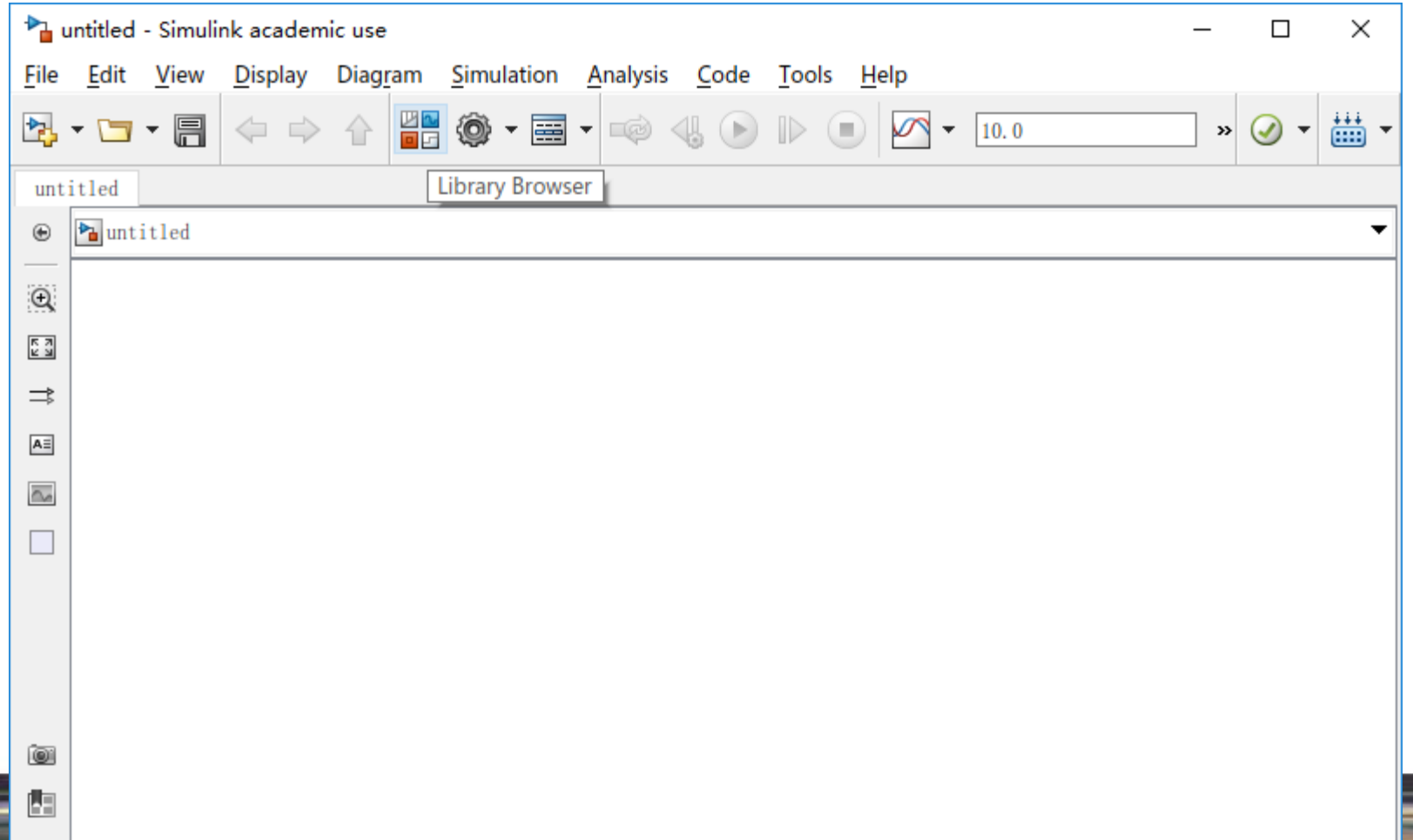
基于simulink的环路仿真

2. 创建blank model



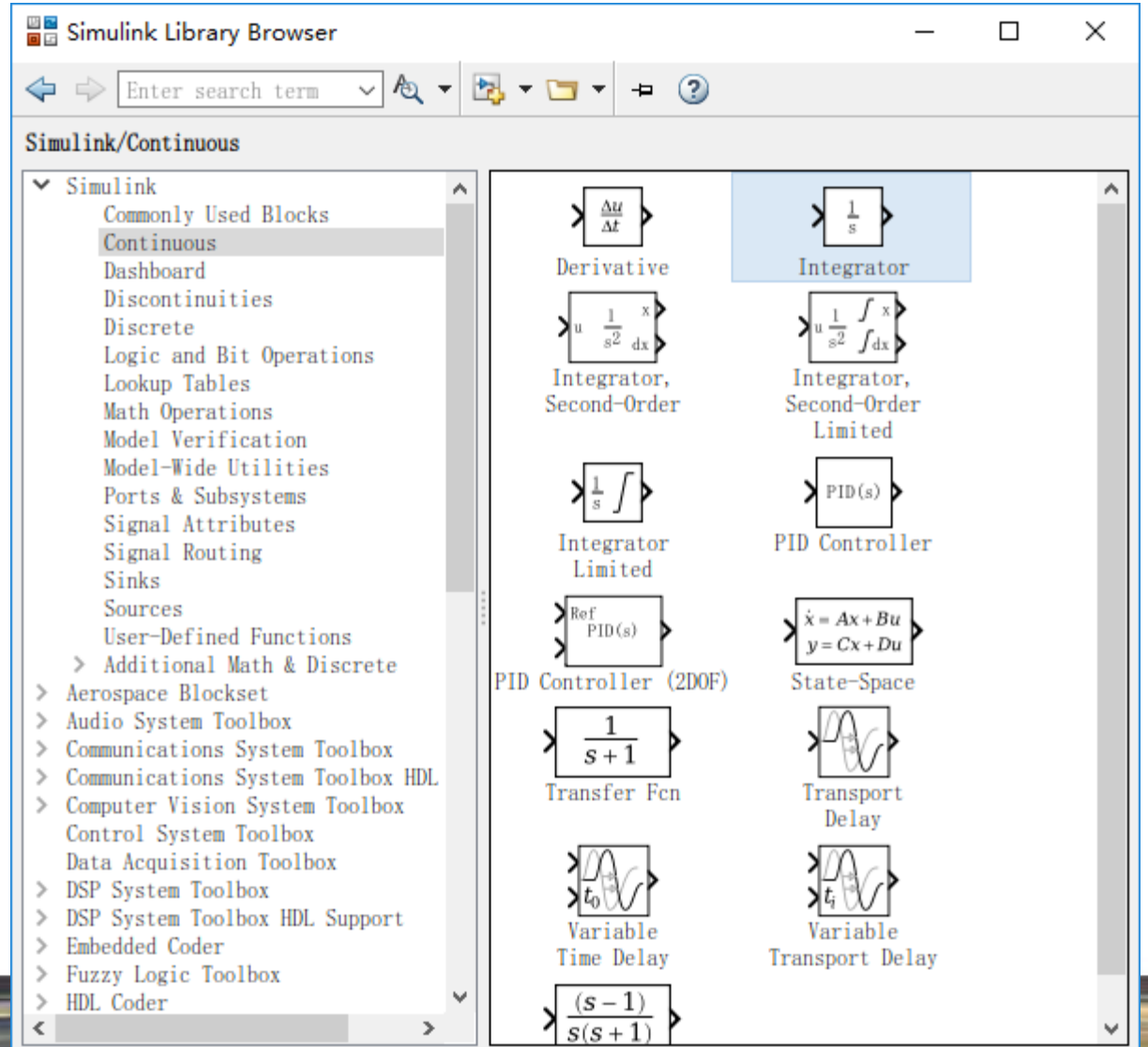
基于simulink的环路仿真

2. 打开Library Browser



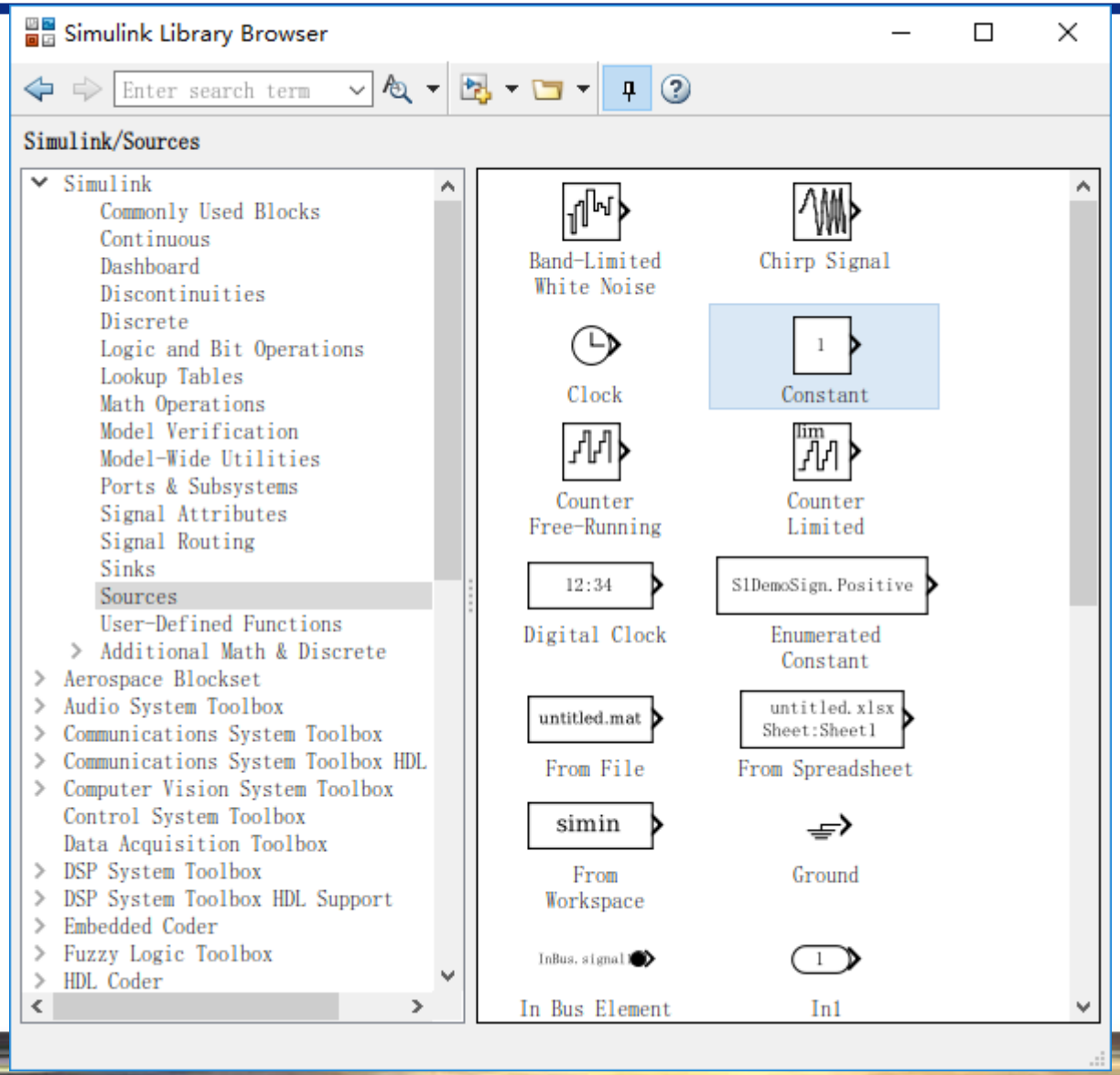
基于simulink的环路仿真

连续系统
仿真模型



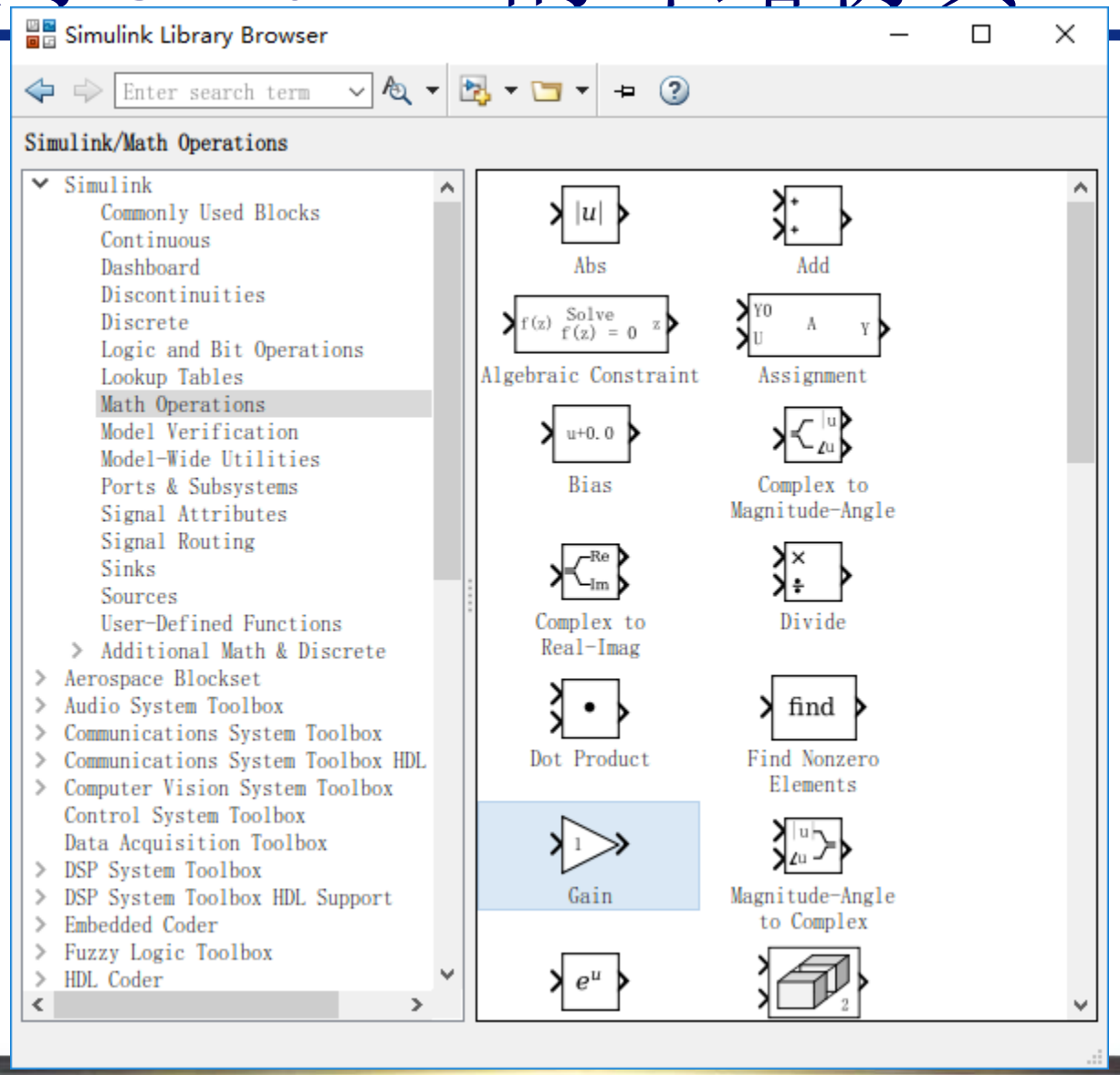
基于simulink的环路仿真

信源



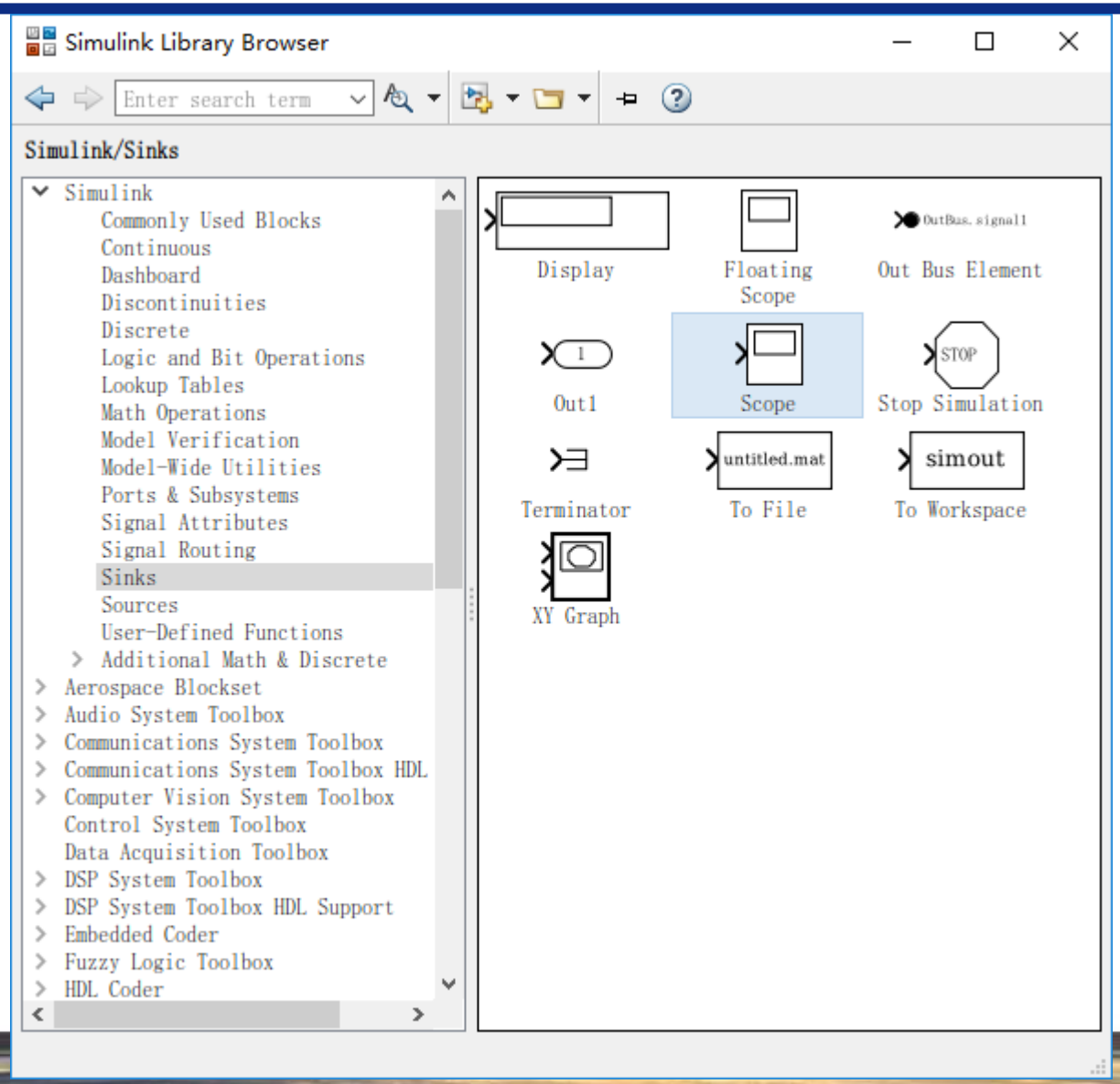
基于simulink的环路仿真

运算模型



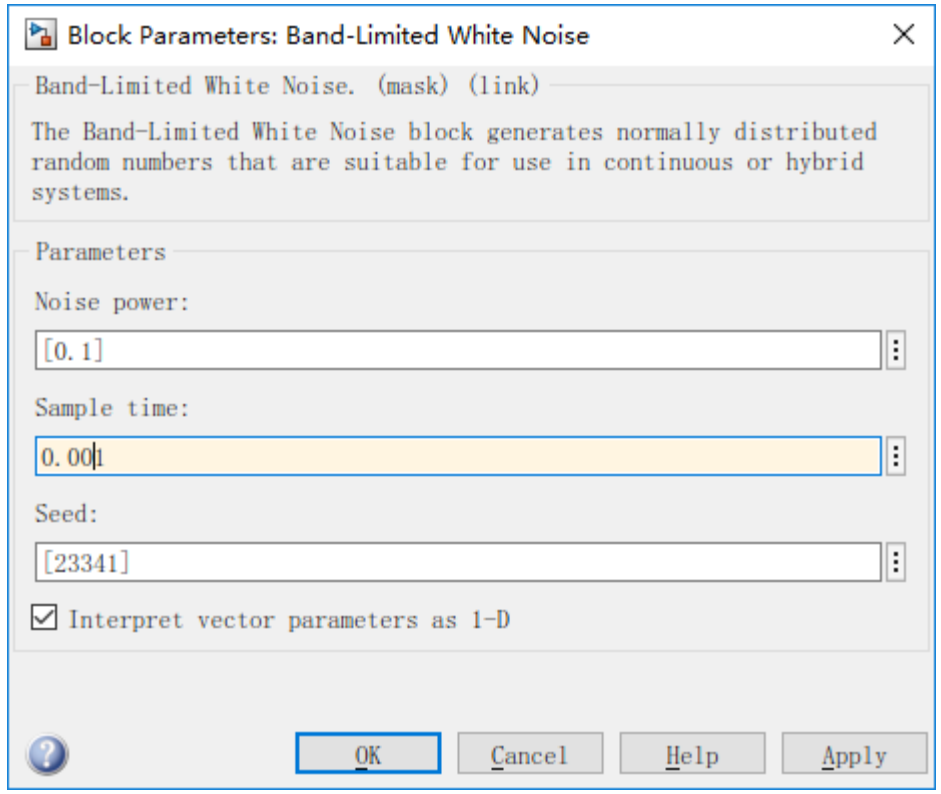
基于simulink的环路仿真

结果输出

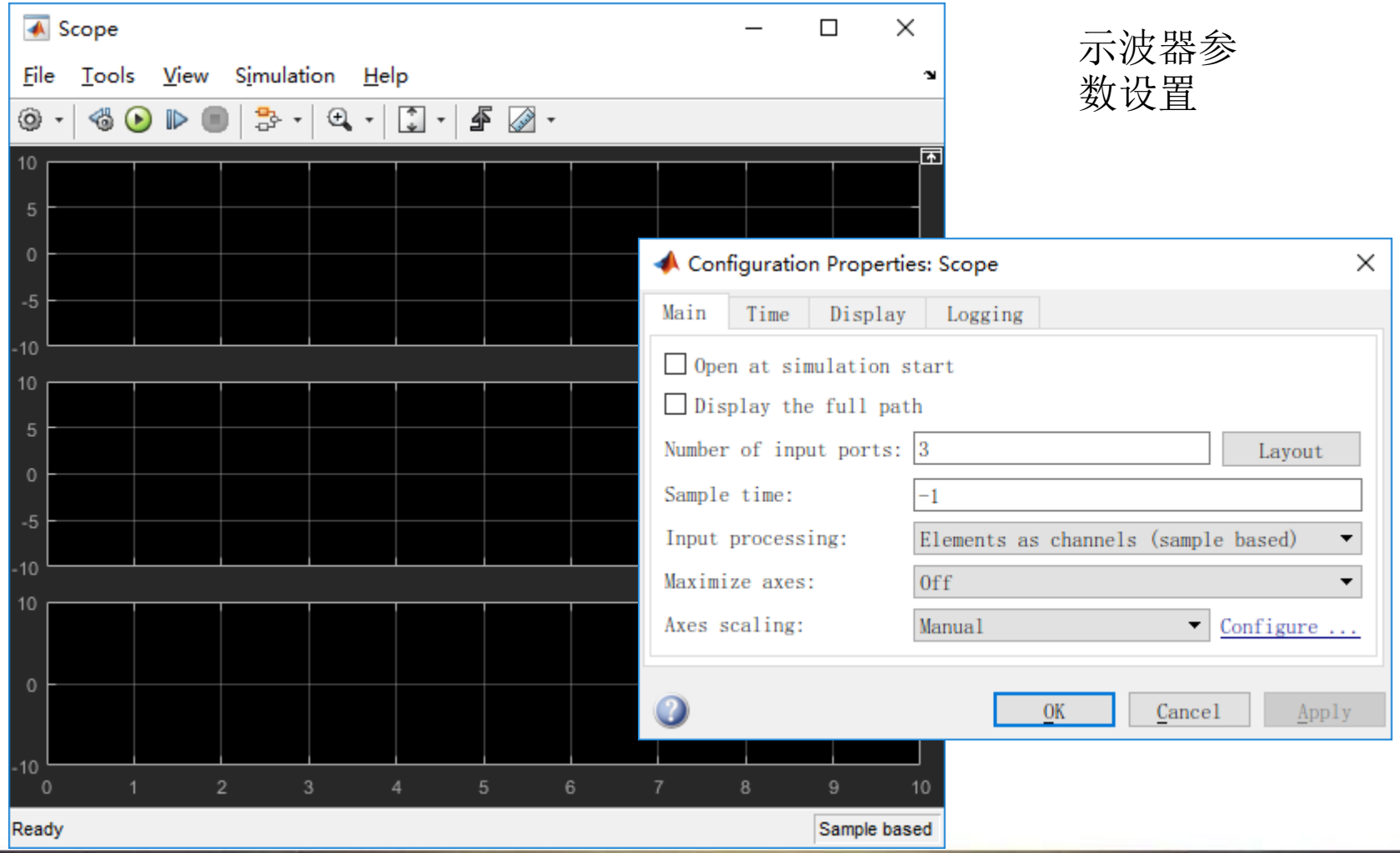


基于simulink的环路仿真

模型参数示例

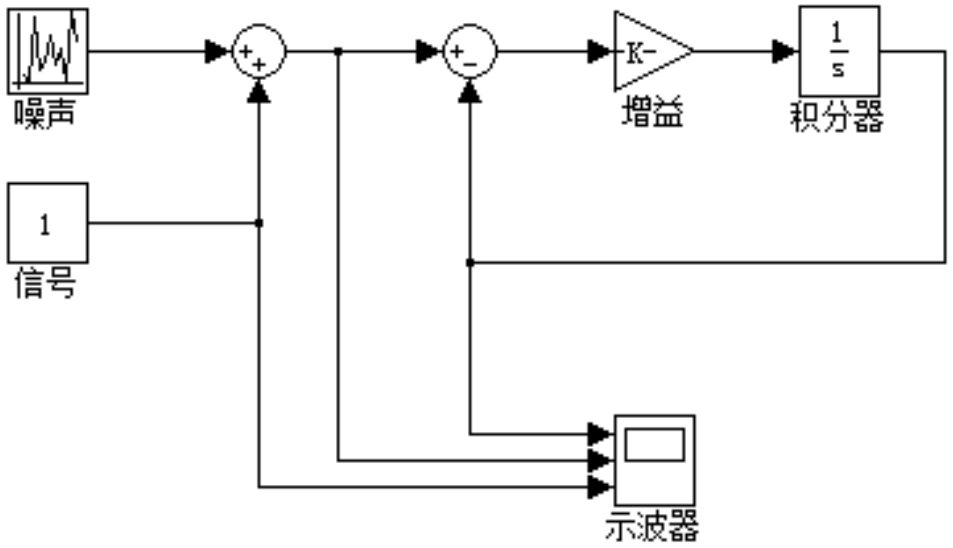


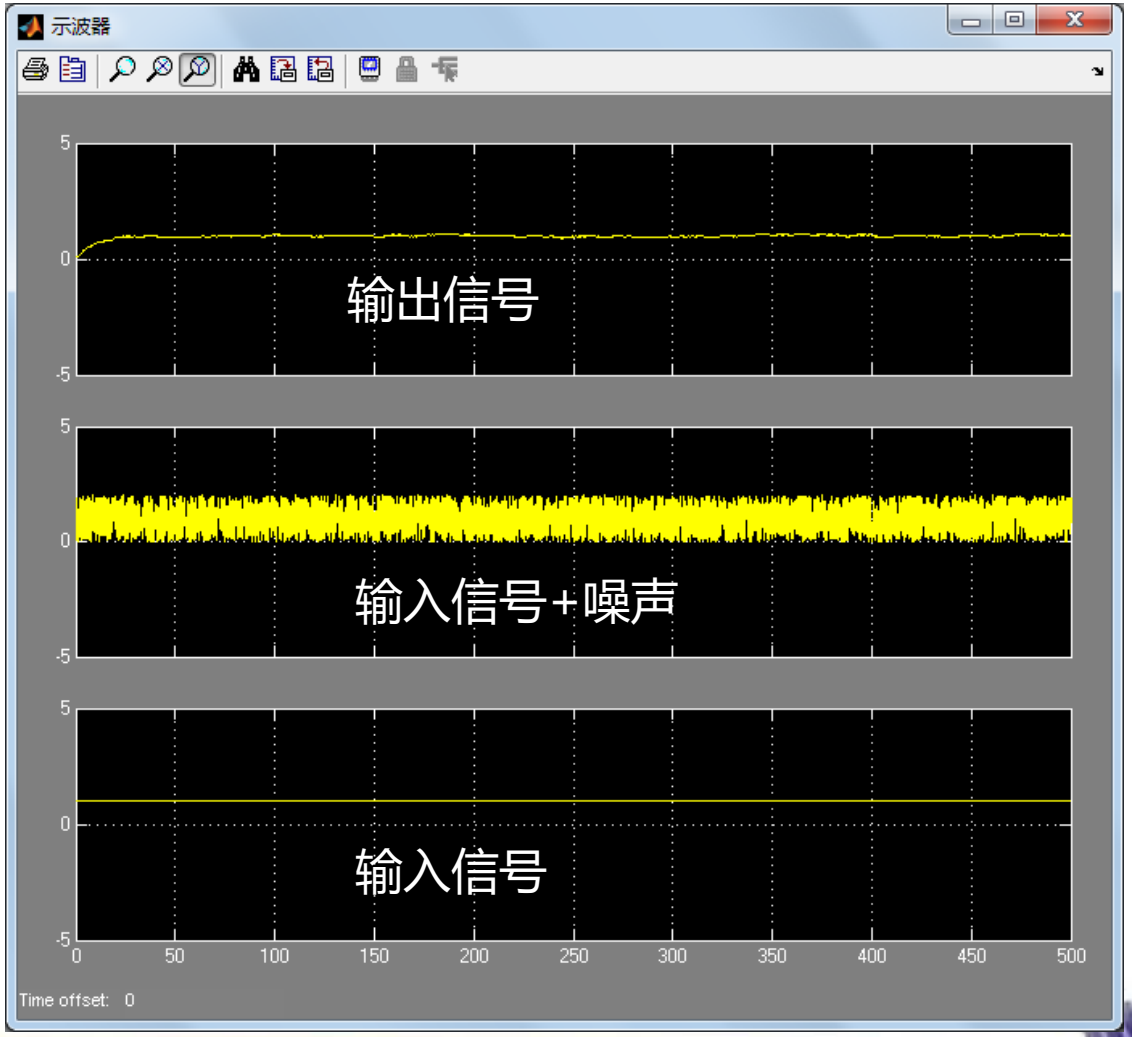
基于simulink的环路仿真



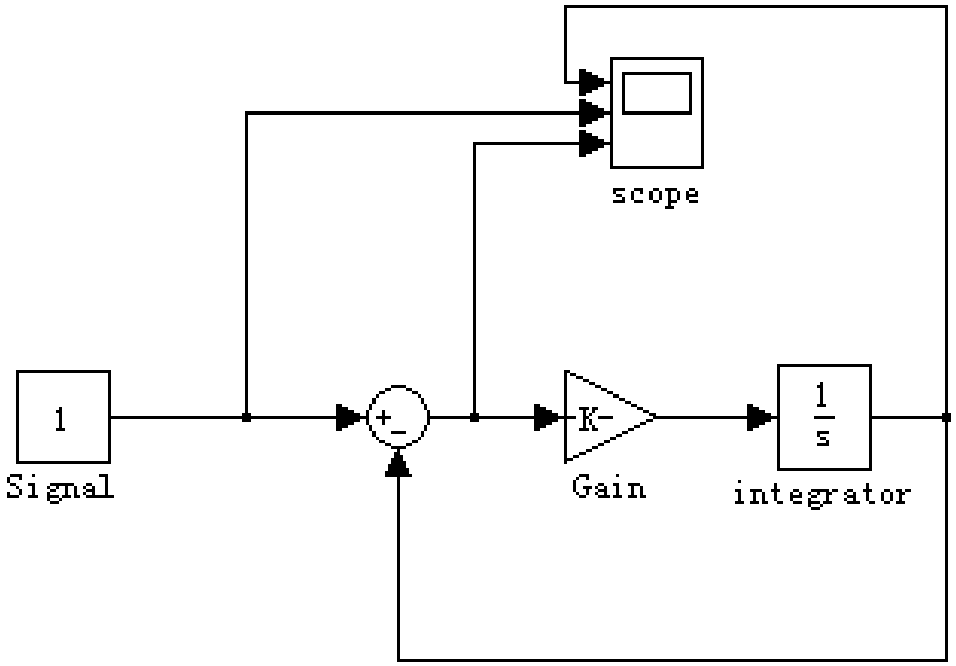
示波器参数设置

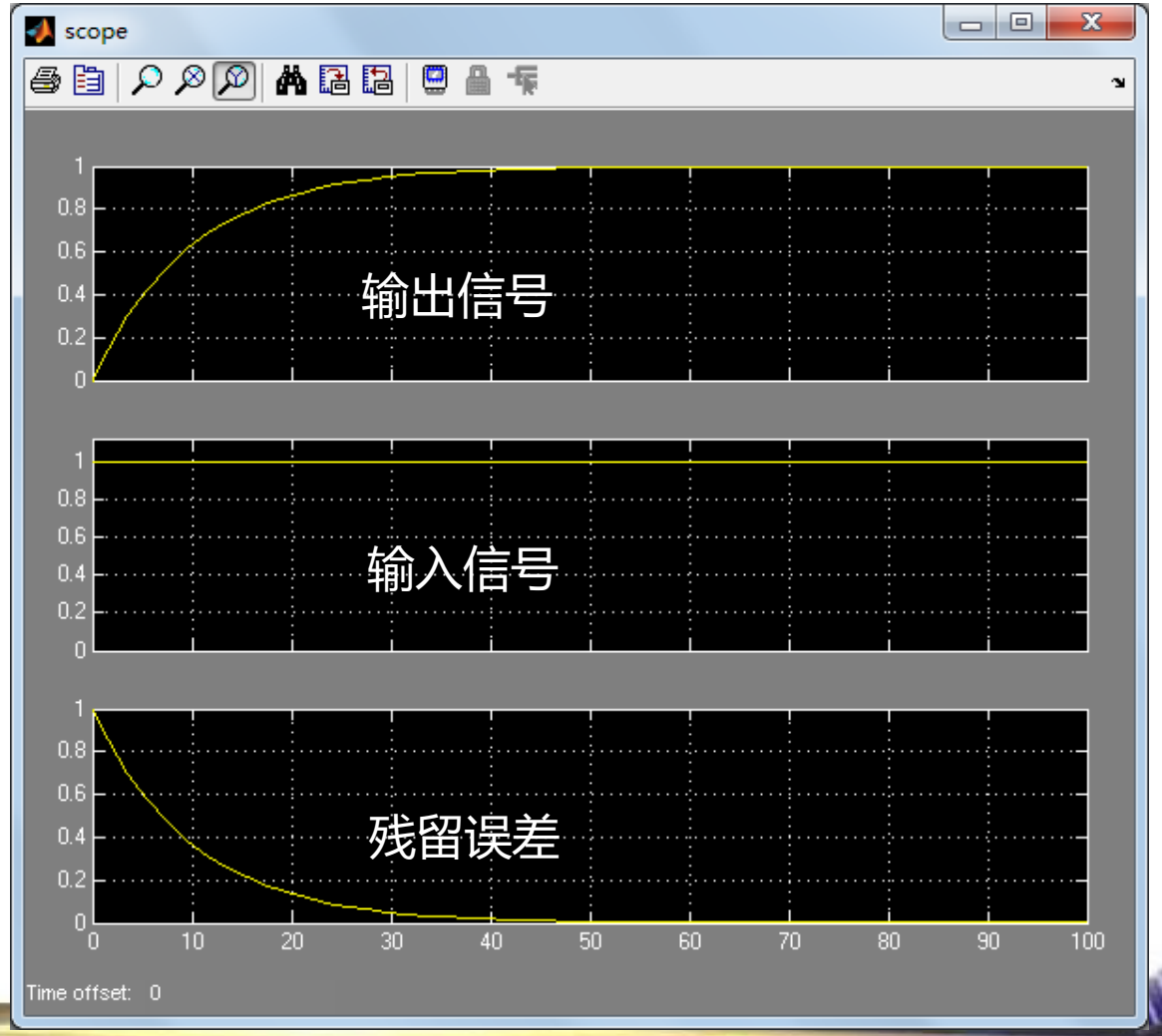
一阶环


噪声
1
信号
增益
K-
1
s
积分器
示波器

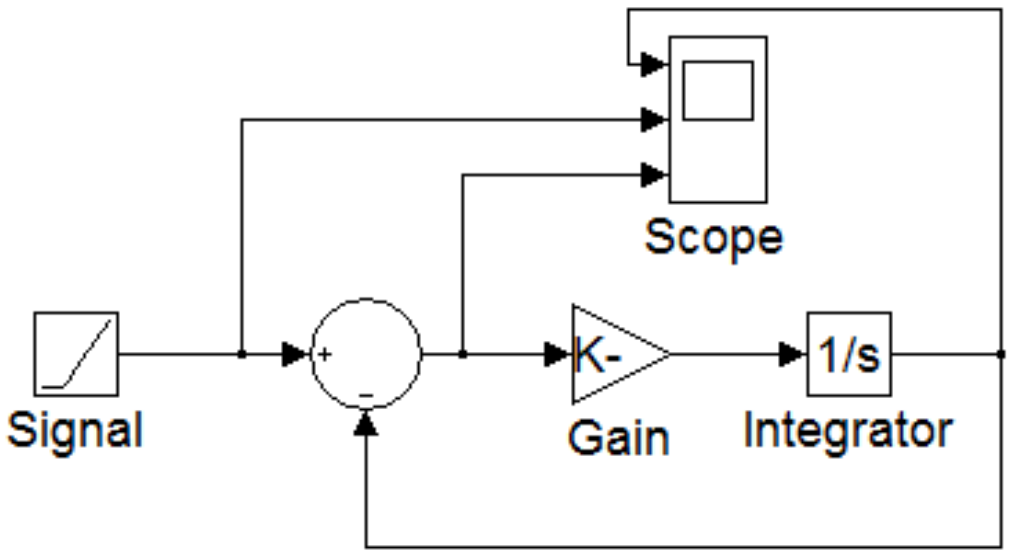

示波器
输出信号
输入信号+噪声
输入信号
5
0
-5
0
50
100
150
200
250
300
350
400
450
500
Time offset: 0

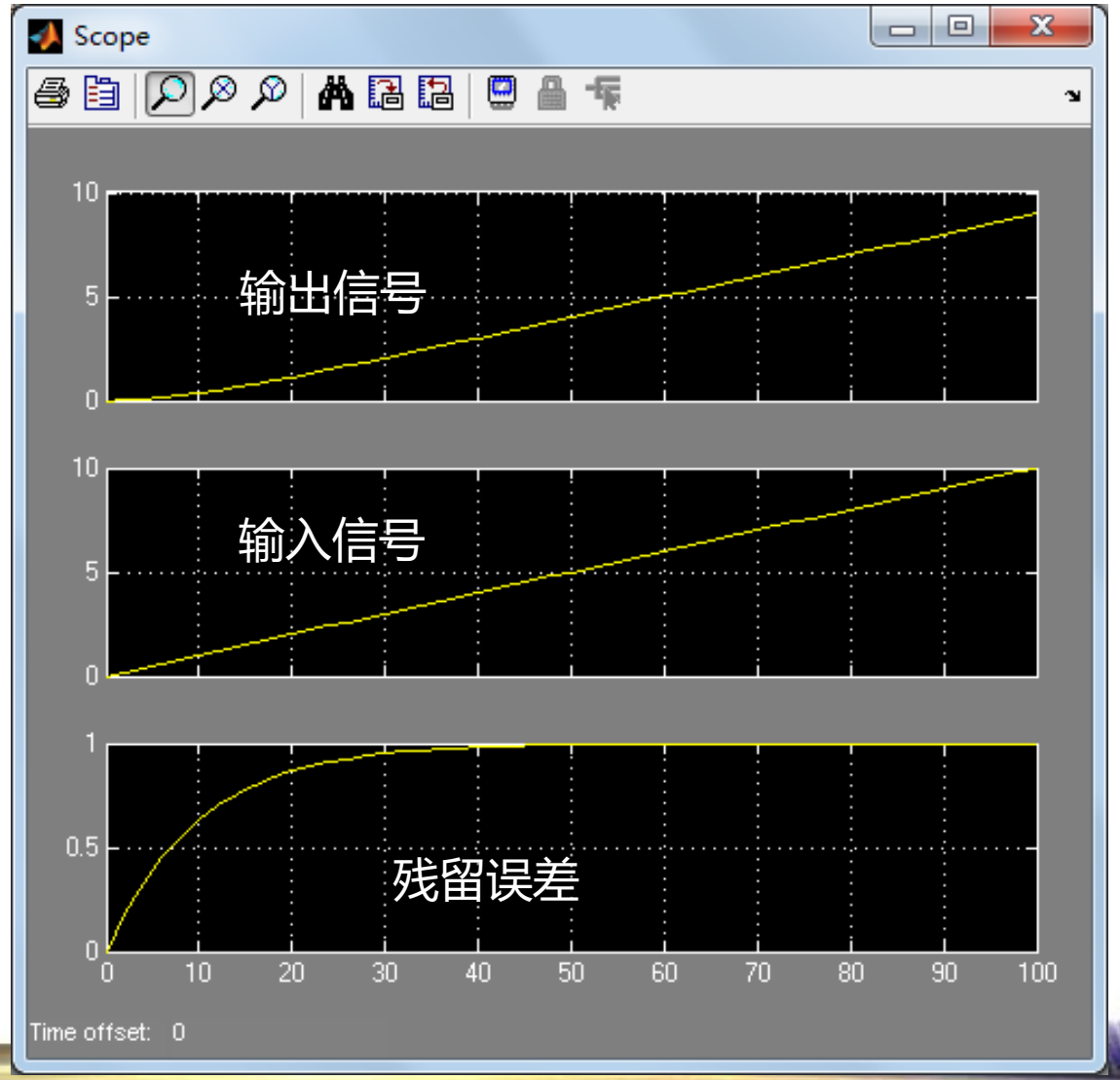
一阶环


1
Signal
scope
K
Gain
1
s
integrator

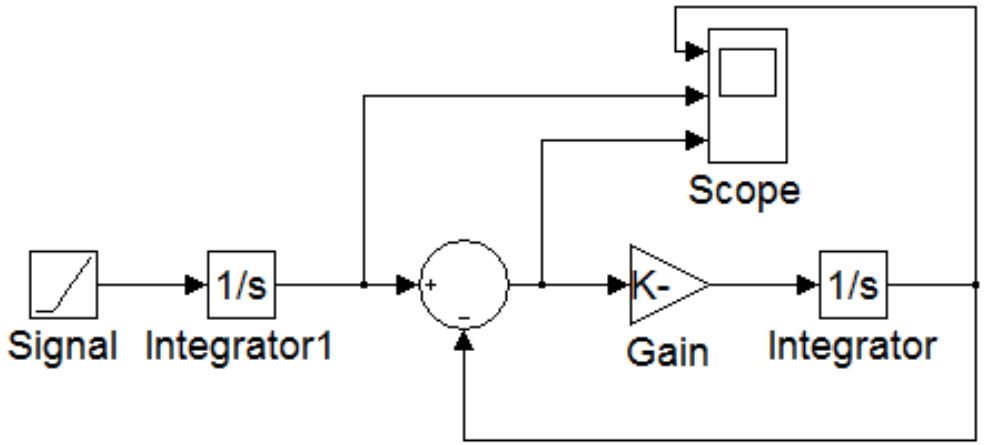

scope
输出信号
输入信号
残留误差
0
0.2
0.4
0.6
0.8
1
0
10
20
30
40
50
60
70
80
90
100
Time offset: 0

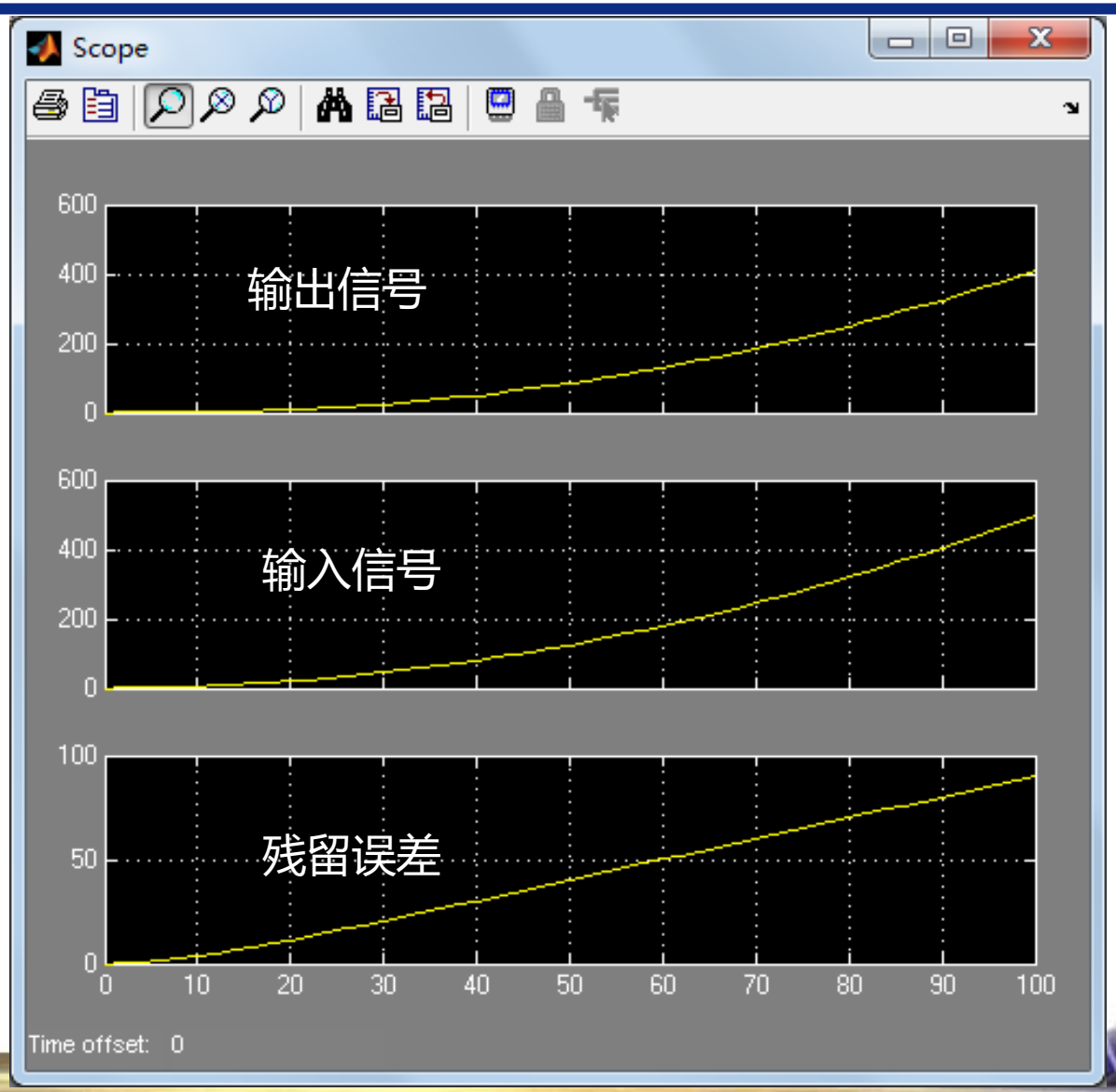
一阶环


Scope
Signal
+
-
K-
Gain
1/s
Integrator

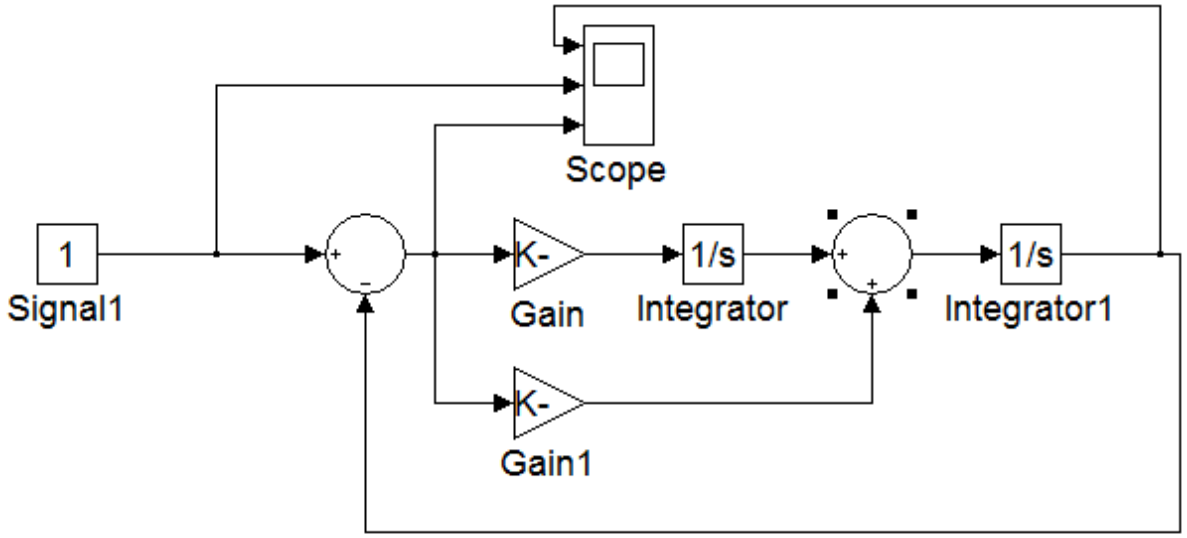

Scope
10
5
0
输出信号
10
5
0
输入信号
1
0.5
0
残留误差
0
10
20
30
40
50
60
70
80
90
100
Time offset: 0

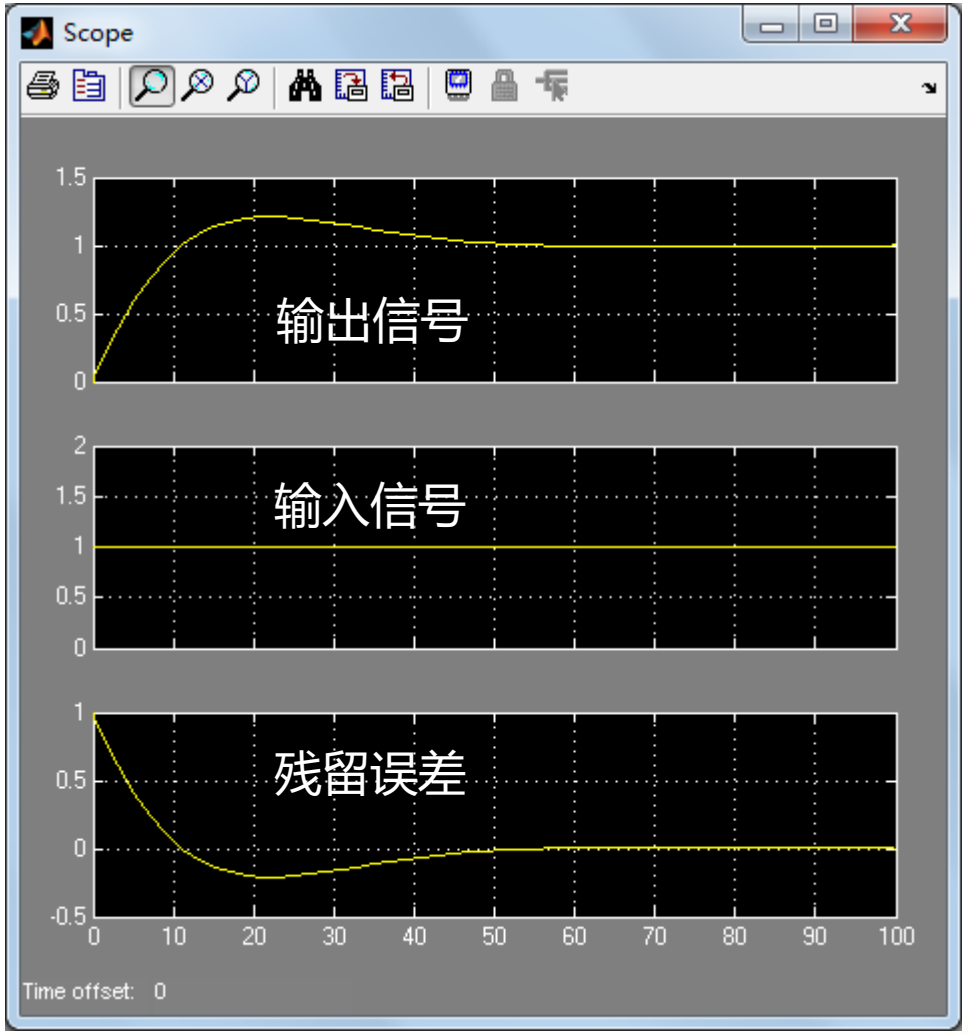
一阶环


Scope
1/s
Signal
Integrator1
K-
Gain
1/s
Integrator

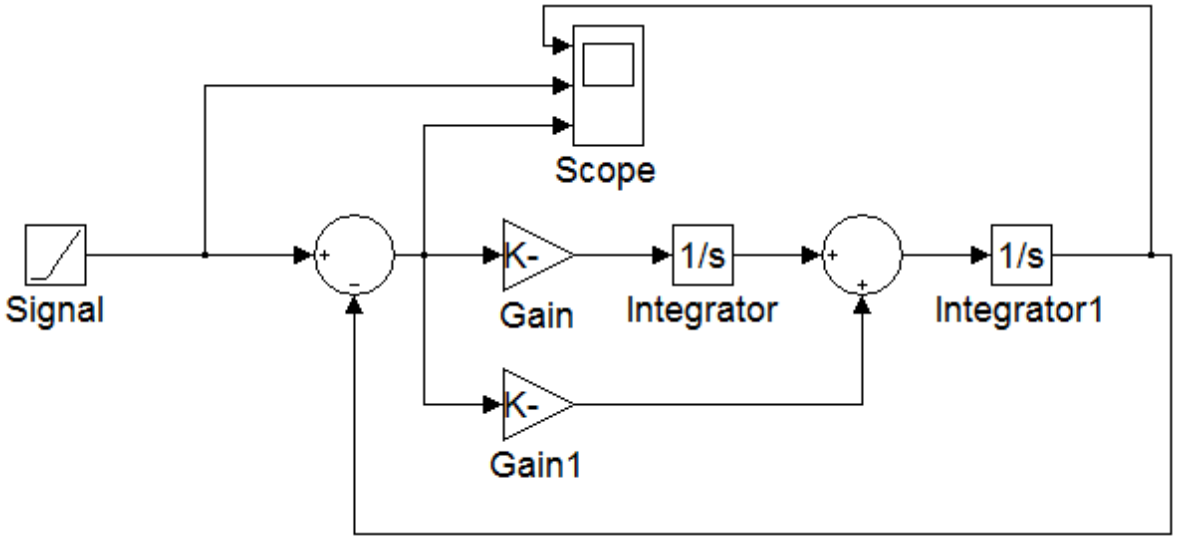

Scope
600
400
200
0
输出信号
600
400
200
0
输入信号
100
50
0
残留误差
0
10
20
30
40
50
60
70
80
90
100
Time offset: 0

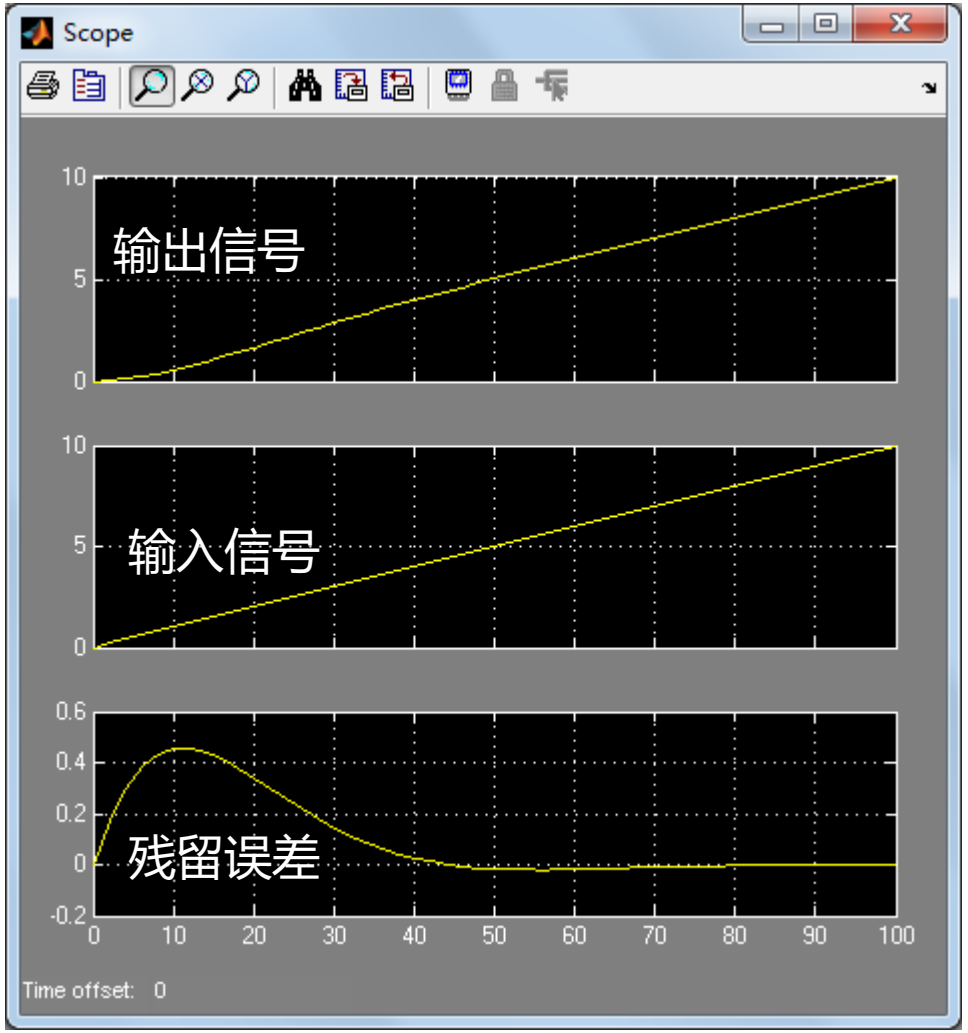
二阶环


Scope
1
Signal1
K-
Gain
1/s
Integrator
1/s
Integrator1
K-
Gain1

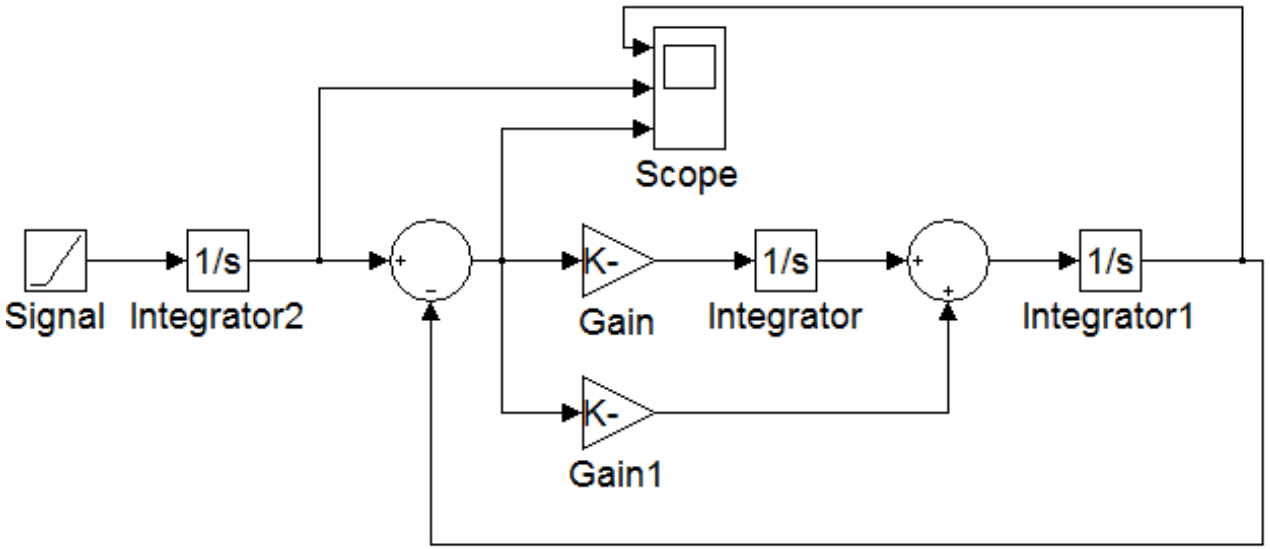

Scope
输出信号
输入信号
残留误差
1.5
1
0.5
0
2
1.5
1
0.5
0
1
0.5
0
-0.5
0
10
20
30
40
50
60
70
80
90
100
Time offset: 0

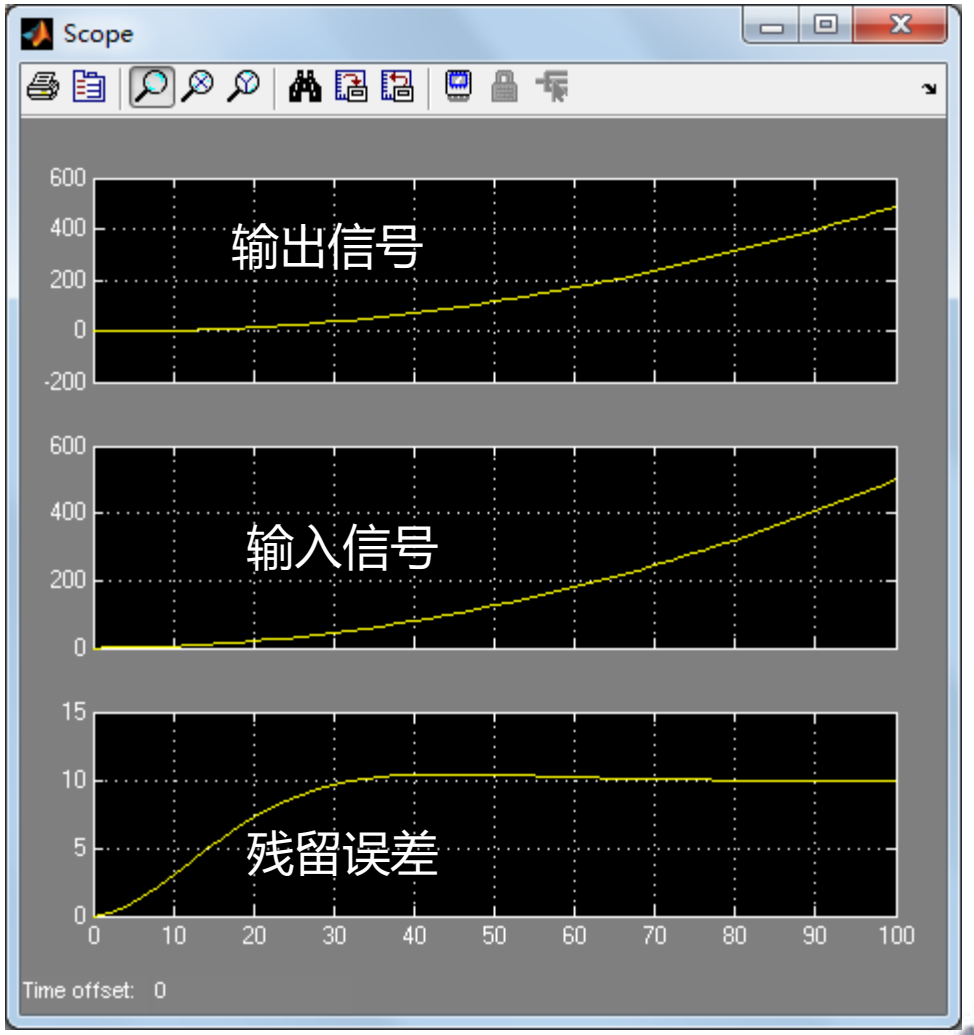
二阶环


Scope
Signal
K-
Gain
1/s
Integrator
1/s
Integrator1
K-
Gain1

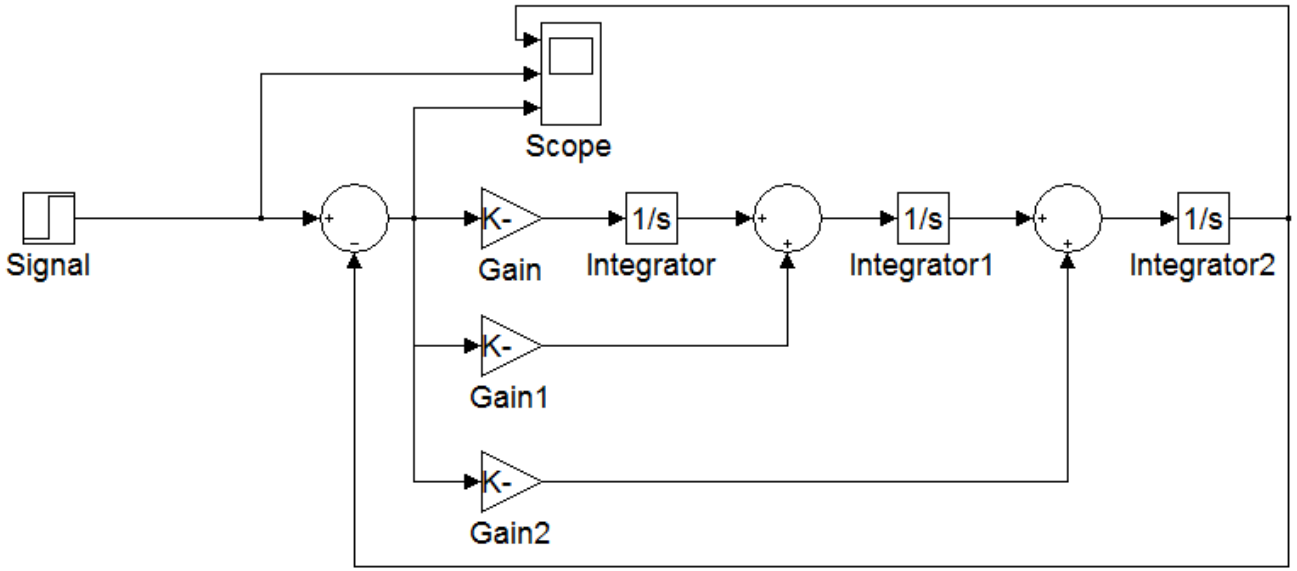

Scope
输出信号
输入信号
残留误差
10
5
0
0.6
0.4
0.2
-0.2
0
10
20
30
40
50
60
70
80
90
100
Time offset: 0

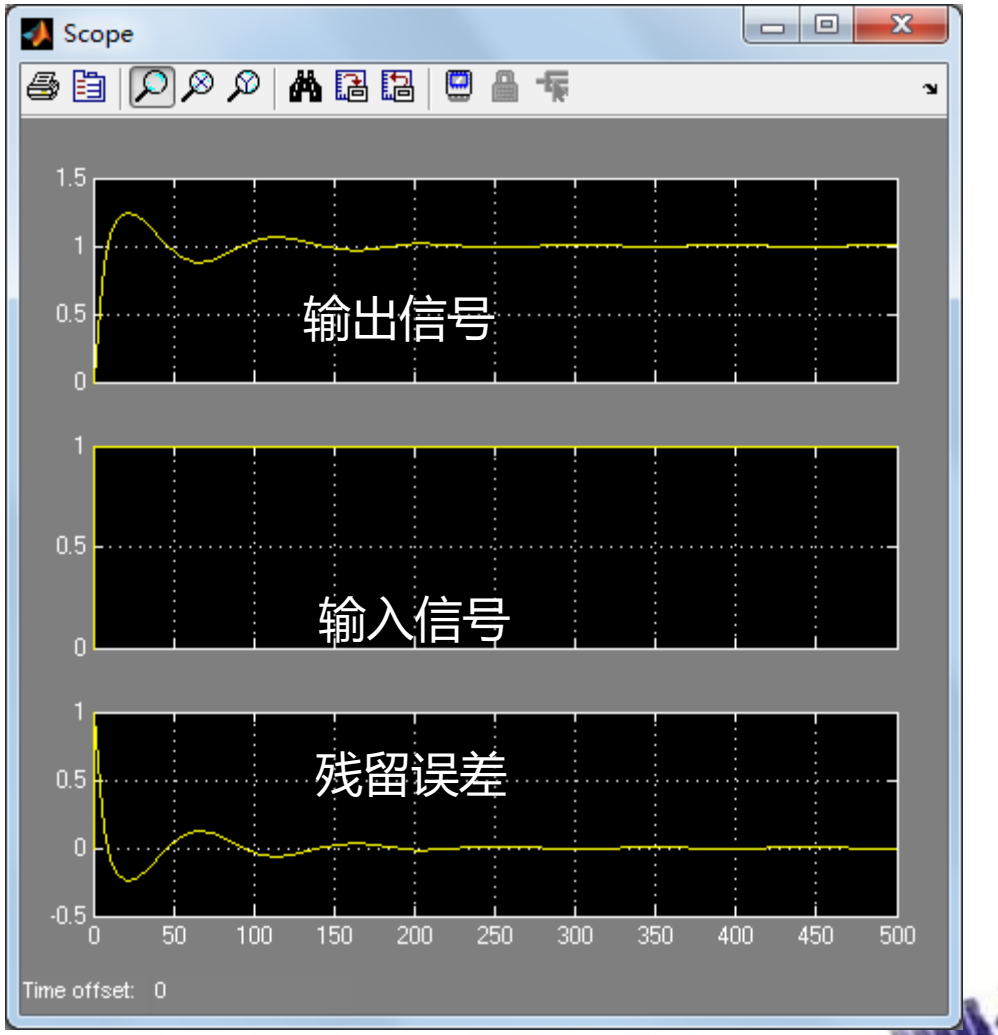
二阶环


Scope
Signal
1/s
Integrator2
K-
Gain
1/s
Integrator
1/s
Integrator1
K-
Gain1

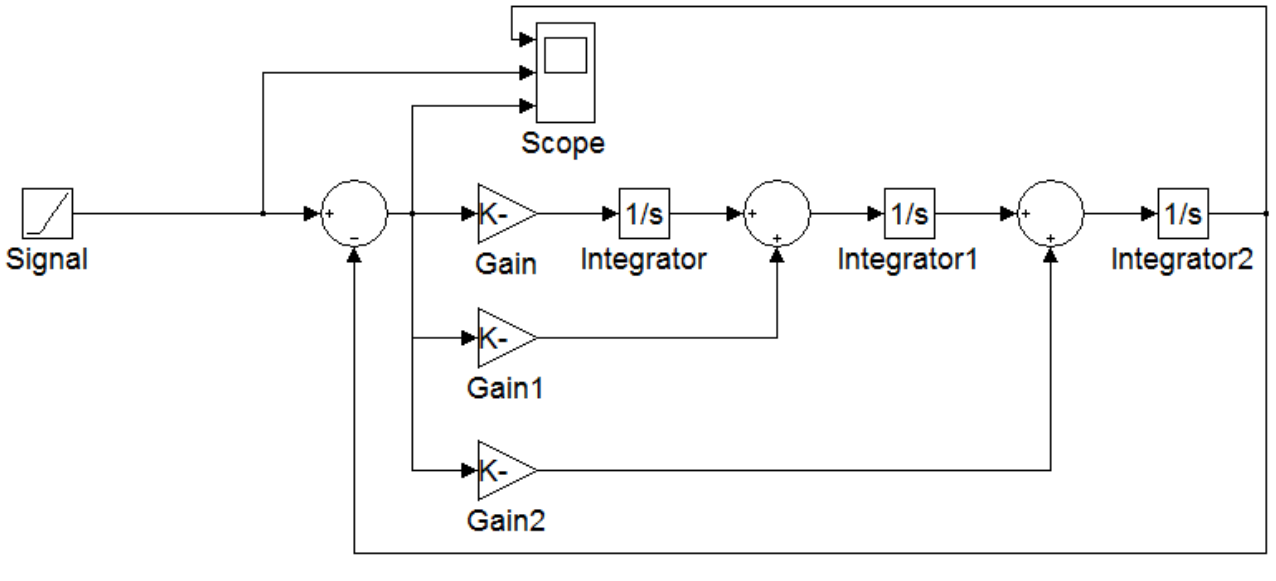

Scope
600
400
200
0
-200
输出信号
600
400
200
0
输入信号
15
10
5
0
残留误差
0
10
20
30
40
50
60
70
80
90
100
Time offset: 0

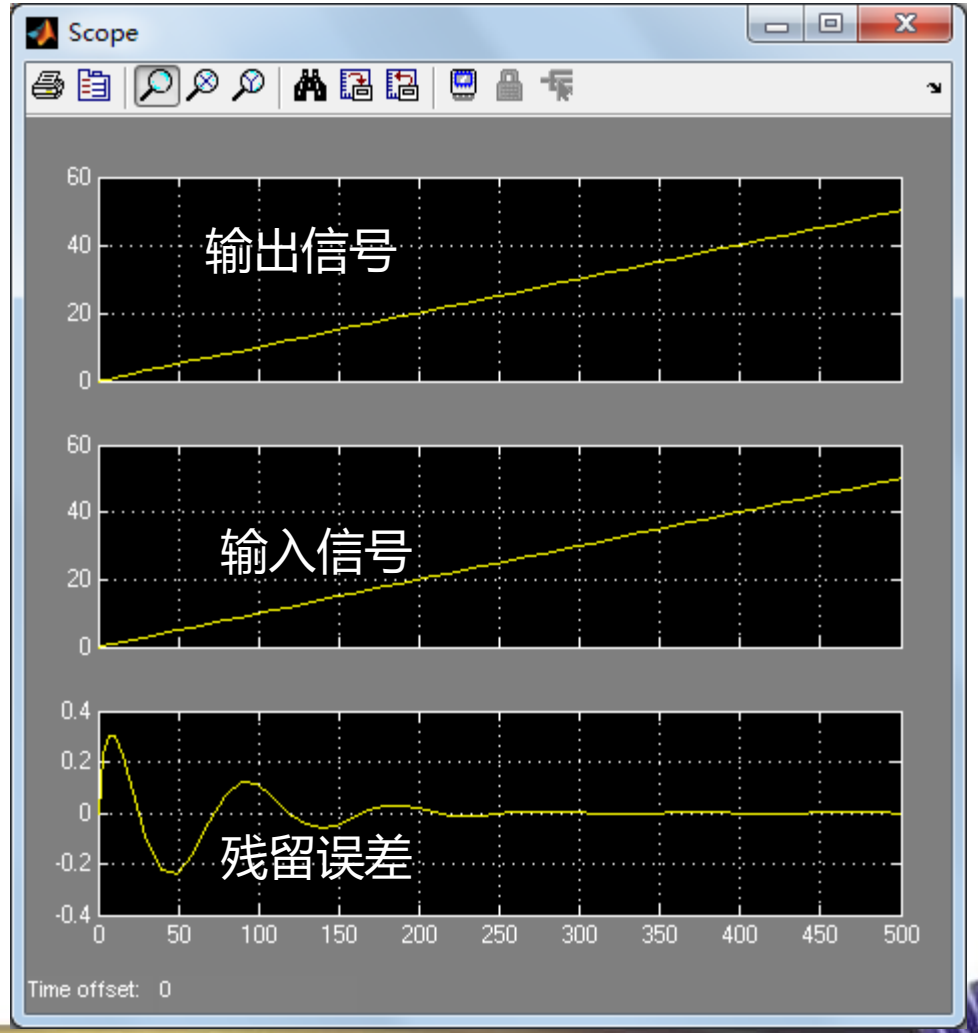
三阶环


Scope
Signal
K-
Gain
1/s
Integrator
1/s
Integrator1
1/s
Integrator2
K-
Gain1
K-
Gain2

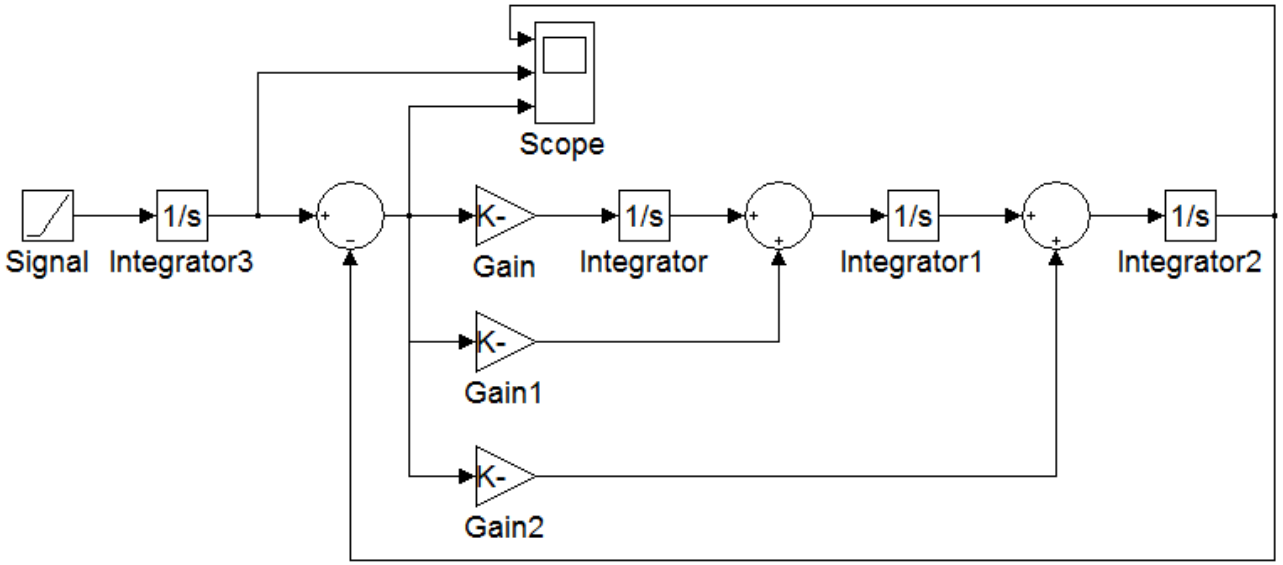

Scope
1.5
1
0.5
0
输出信号
1
0.5
0
输入信号
1
0.5
0
-0.5
残留误差
0
50
100
150
200
250
300
350
400
450
500
Time offset: 0

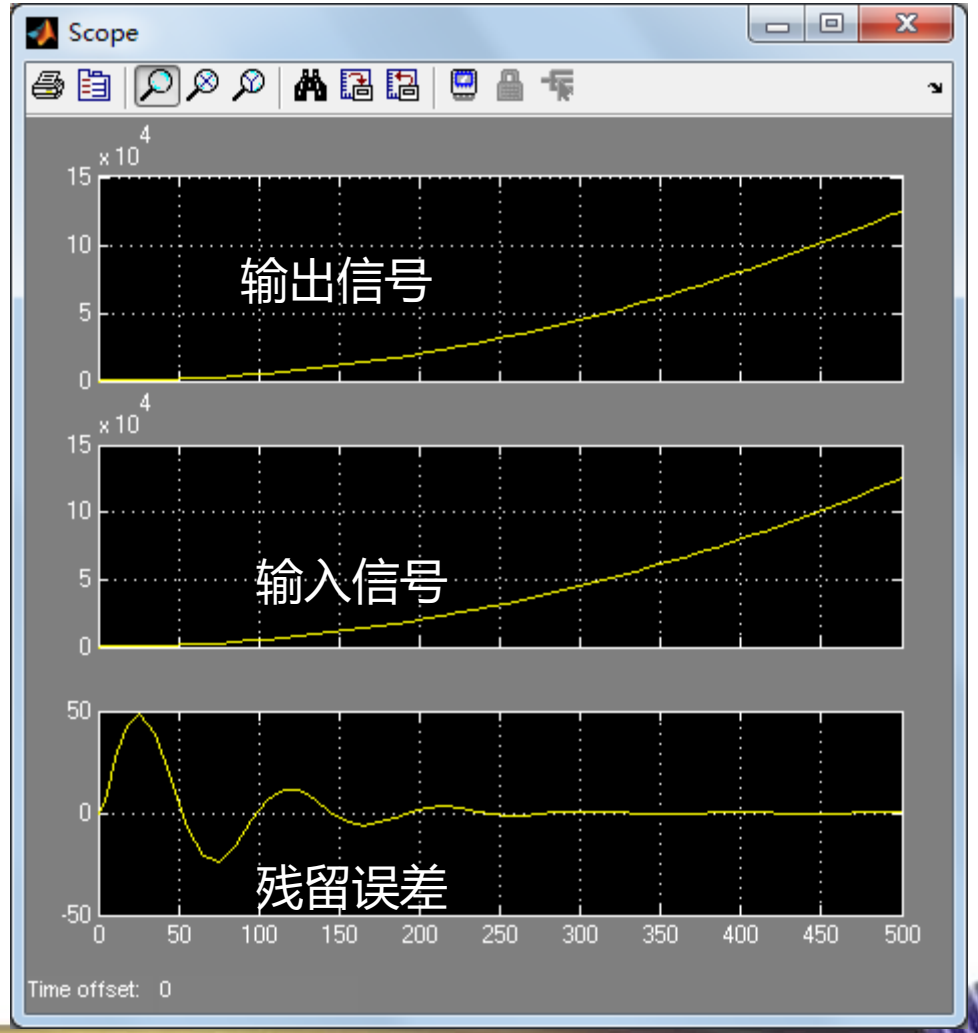
三阶环


Scope
Signal
K-
Gain
1/s
Integrator
1/s
Integrator1
1/s
Integrator2
K-
Gain1
K-
Gain2


Scope
60
40
20
0
输出信号
60
40
20
0
输入信号
0.4
0.2
0
-0.2
-0.4
残留误差
0
50
100
150
200
250
300
350
400
450
500
Time offset: 0

三阶环


Scope
Signal
1/s
Integrator3
K-
Gain
1/s
Integrator
1/s
Integrator1
1/s
Integrator2
K-
Gain1
K-
Gain2


Scope
x 10^4
15
10
5
0
输出信号
x 10^4
15
10
5
0
输入信号
50
0
-50
残留误差
0
50
100
150
200
250
300
350
400
450
500
Time offset: 0

实验内容

❑ 仿真并记录

- 一阶、二阶、三阶环路对阶跃、斜坡、匀加速信号的响应
- 环路带宽对跟踪稳定时间的影响（以阶跃响应为例）

环路带宽	一阶环	二阶环	三阶环
2Hz			
1Hz	跟踪稳定时间		
0.1Hz			

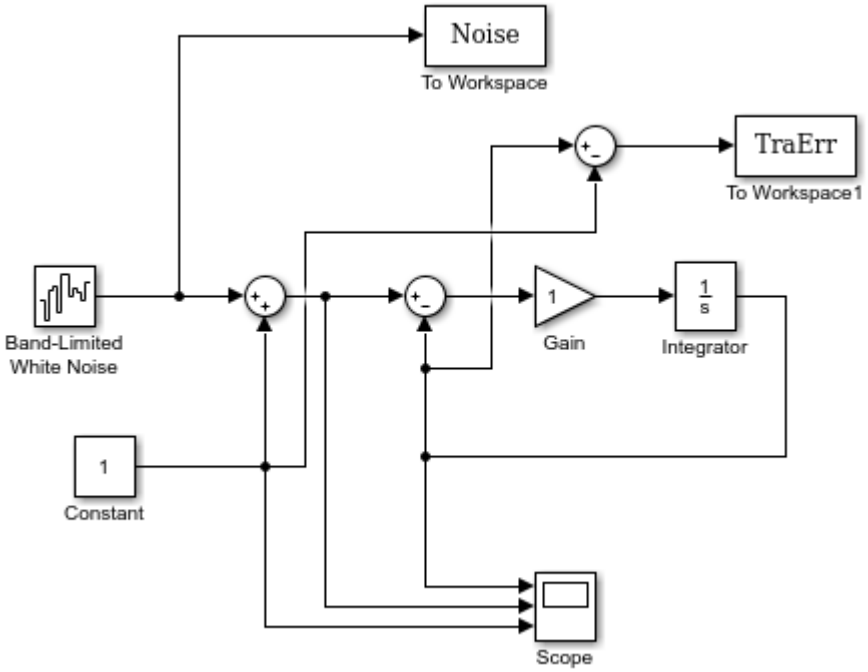
- 环路带宽对噪声抑制性能的影响
  - 以一阶环和二阶环为例。将仿真结果输出到**workspace**，存储在设定的变量中，并统计噪声功率和均方根误差。（输入噪声双边功率谱为**1**）

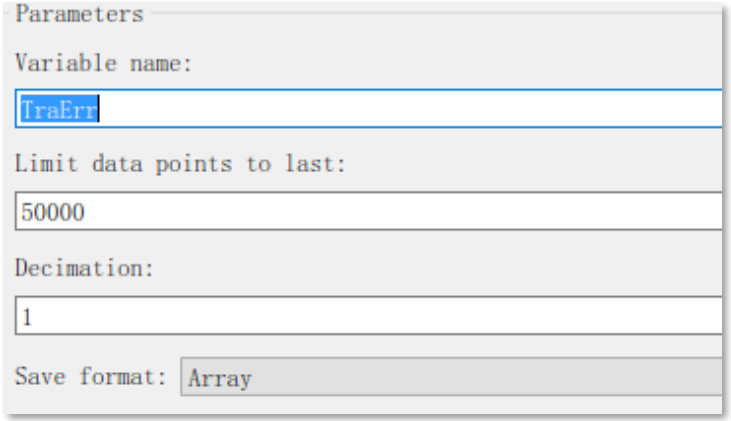
环路带宽	一阶环	二阶环
1Hz		
0.1Hz		

实验内容

❑ 仿真并记录

- 环路带宽对噪声抑制性能的影响
  - 以一阶环和二阶环为例，统计功率
    可将仿真结果输出到**workspace**，
    并统计噪声功率和均方根误差





双边功率谱密度：1

采样间隔： 0.01s
环路带宽：1/4Hz
（单边）

```
>> rms(Noise).^2

ans =

   100.1658

>> rms(TraErr).^2

ans =

    0.4819
```